

Data-Centric Challenges, Techniques, and Impacts: A Survey on Image Data Perturbation

Peng-Fei Zhang , Guangdong Bai , Xin-Shun Xu , *Senior Member, IEEE*, and Zi Huang , *Fellow, IEEE*

(Survey Paper)

Abstract—Data-driven deep learning models have revolutionized the ability to understand and model intrinsic data patterns. However, in real-world applications, their reliability is often compromised by inherent randomness, uncertainty, and potential adversarial threats, particularly those originating from data. As a critical component of data engineering, data perturbation has emerged as an effective approach for evaluating and enhancing the robustness of deep learning models, encompassing both inadvertent and deliberate modifications to data. This survey offers a comprehensive review of data perturbations, with a particular focus on the extensively studied image perturbations on classification throughout the development and deployment of deep models. It presents a detailed summary and taxonomy of existing perturbations, their generation methods, interrelationships, implications for model robustness, and explores underlying mechanisms. By systematically analyzing recent advances and best practices across both digital and physical domains, our goal is to provide an in-depth understanding of how data engineering, which is particularly through data perturbation, can be leveraged to develop models that are both accurate and resilient. Additionally, we discuss current limitations in current research and suggest promising directions for future study.

Index Terms—Image perturbations, adversarial attacks, defense, deep neural networks, robust learning.

I. INTRODUCTION

THE development of deep learning models has significantly advanced the understanding, representation, and organization of data, thereby enhancing the effectiveness of data-driven models across diverse tasks. However, the effectiveness, reliability, and robustness of these models are constrained and influenced by the complex nature of data, particularly due to inherent randomness, uncertainty, and adversarial threats in real-world environments. These challenges highlight the importance of robust data engineering practices, particularly data perturbation techniques, which play a critical role in stress-testing models, uncovering vulnerabilities, and improving their generalizability and resilience in real-world scenarios.

Received 27 August 2025; revised 24 May 2026; accepted 26 June 2026. Date of current version 8 August 2026. This work was supported by the ARC under Grant DP230101196. Recommended for acceptance by Y. Tong. (Corresponding author: Zi Huang.)

Peng-Fei Zhang, Guangdong Bai, and Zi Huang are with the School of Electrical Engineering and Computer Science, The University of Queensland, Brisbane, QLD 4072, Australia (e-mail: mima.zpf@gmail.com; g.bai@uq.edu.au; huang@itee.uq.edu.au).

Xin-Shun Xu is with the School of Software, Shandong University, Jinan 250100, China (e-mail: xuxinshun@sdu.edu.cn).

This article has supplementary downloadable material available at <https://doi.org/10.1109/TKDE.2026.3709786>, provided by the authors.

Digital Object Identifier 10.1109/TKDE.2026.3709786

Data, as a foundational element, permeates every aspect of deep learning models. During training, it serves as the primary resource, enabling models to capture data content and structure and to develop robust representation capabilities. Once trained, deep learning models are deployed to make predictions on testing data. Essentially, the success of deep learning models relies heavily on the availability of representative data. Data perturbations are most common and can significantly impact the quality and effectiveness of data. Therefore, studying model robustness through the lens of data perturbations is a viable and important approach.

The study of data perturbations and their impact on model robustness has a long history. Perturbations are widely observed in various tasks, e.g., classification [1], [2], [3], [4], and retrieval [5], [6], [7], and data forms, e.g., image [1], [5], video [8], [9], audio [10], [11], and text [12], [13]. Specifically, data perturbations influence model robustness in two significant ways. On the one hand, perturbations can cause models to misinterpret inputs by affecting the inference (e.g., evasion attack [1], [2], [5], [14] and even training phases (e.g., poisoning attack [15], [16], [17], [18], thereby impacting model performance. On the other hand, perturbations can also be leveraged to enhance model robustness during both training (e.g., via adversarial training [19], [20], [21], [22], and data augmentation [23], [24], [25], [26]) and test time (e.g., via randomized smoothing [27], [28], [29], [30]).

Given the extensive research on data perturbations, a comprehensive view is necessary to fully grasp their influence and nature. However, existing surveys related to data perturbations are fragmented and task-specific, focusing on areas like evasion attack and defense [39], [40], poisoning attack [41], backdoor attack [42], and specific models [43]. A systematic and unified understanding of data perturbations across diverse attacks, defenses, model architectures and tasks/scenarios is still lacking. This includes categorisation of perturbation types, exploring cross-perspectives between different perturbations and their enabled attacks and defenses, understanding the root causes of model vulnerabilities and robustness, comparing behaviors across architectures, and addressing current challenges in perturbation-based attacks and defenses. To fill this gap, we present a survey on data perturbations covering various deep models, including convolutional neural networks [32], [44], vision transformers [45], [46], diffusion models [47], [48], [49], vision-language models [50], [51], and large language

TABLE I
TAXONOMY AND DESCRIPTION OF IMAGE PERTURBATIONS

Mechanism	Perturbation Type	Level	Description
Pixel-level (signal) perturbations	Noise and blur	Low	Adversarial noise, Gaussian noise, shot noise, impulse noise, defocus blur, motion blur, and zoom blur [31].
	Colour perturbations	Low	Intensity, hue, saturation, and colour space transformations [32], [33].
	Texture perturbations	Low–Mid	Local surface patterns and fine-grained visual details [34], [35].
	Compression artifacts	Low	JPEG compression, pixelation, and quantization effects [36].
Geometric (spatial) perturbations	–	Mid	Rotation, translation, scaling, skewing, flipping, cropping, and warping [31], [25].
Global appearance perturbations	Style perturbations	High	Modify global appearance statistics such as colour, contrast, and texture distribution [23].
Content-level perturbations	Subtractive perturbations	High	Remove regions or objects (e.g., occlusion or masking) [24].
	Mix-based perturbations	High	Combine multiple images or inject external content into the original data [37], [38].
Environmental perturbations	Weather perturbations	High	Simulate fog, rain, snow, and illumination changes, which manifest through noise, blur, and contrast degradation [31].

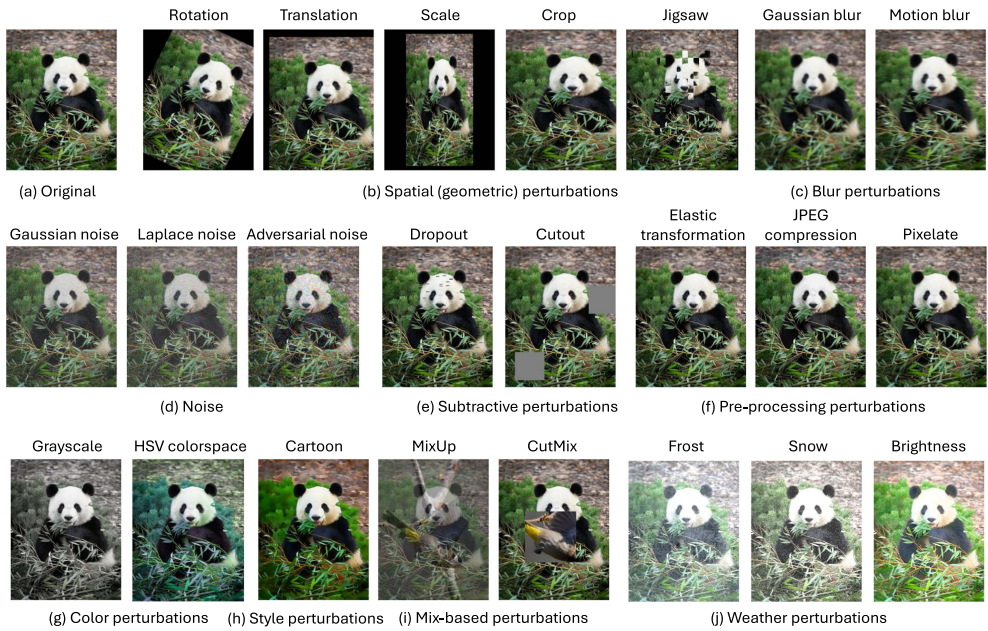


Fig. 1. Examples of images affected by different perturbations, where high-severity perturbations are used for better illustration.

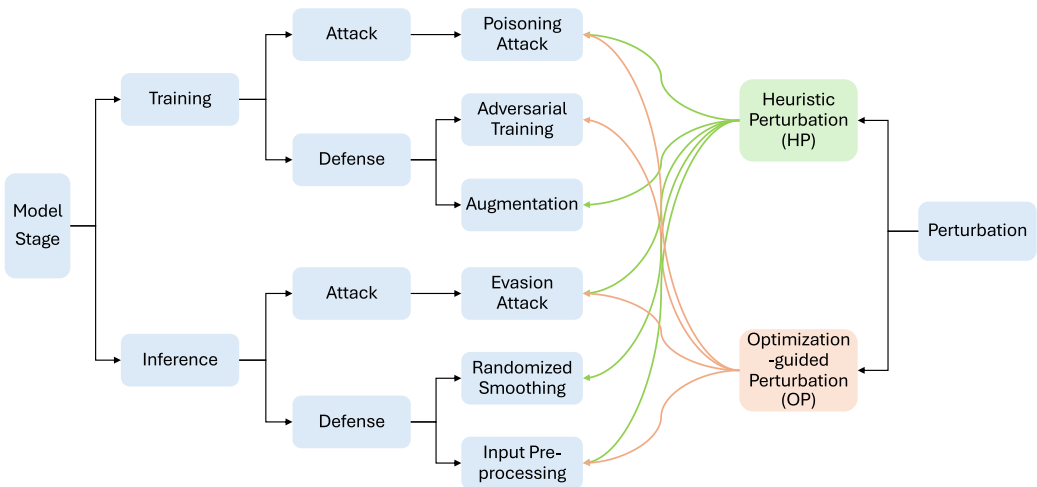


Fig. 2. Applications of image perturbations. Perturbations, including heuristic and optimization-guided perturbations, are widely used across various techniques aimed at evaluating and improving AI reliability.

models [52], [53]. For clarity and focus, we primarily center our discussion on image perturbations on classification, which have received the most attention and whose underlying designing principles are broadly applicable to other data types. Our survey investigates various types of perturbations (Table I and

Fig. 1), their applications, interconnections and impacts on model robustness in both the digital and physical world. We offer cross-perspectives across different tasks and architectures, and highlight key challenges and existing solutions, including adversarial transferability, efficiency, imperceptibility, multi-attack

TABLE II
APPLICATION DESCRIPTION OF IMAGE PERTURBATIONS

Stage	Description		
Training	Attack	Poisoning attack	Corrupt the training process of DNNs by injecting perturbations into either a portion of or the entire training dataset with the aim of influencing models' behaviour during testing.
	Defense	Adversarial training	Generate worst-case adversarial perturbations into training data to enhance model robustness.
		Augmentation	Generate non-adversarial perturbations into training data to enhance model robustness
Inference	Attack	Evasion attack	Apply perturbations to input data during the evaluation or deployment phase to degrade the performance of DNNs
	Defense	Randomized smoothing	Transform a base classifier into a "smoothed" classifier by injecting random noise into the input data and averaging the predictions over multiple noisy samples
		Input pre-processing	Employ specific perturbation operations as a preprocessing step to disrupt the perturbation patterns.

TABLE III
CHALLENGE-ORIENTED COMPARISON OF REPRESENTATIVE PERTURBATION STRATEGIES. ✓: EXPLORED; ○: NOT EXPLICITLY STUDIED; —: NOT APPLICABLE.

Challenge Dimension	HP-based Evasion Attack	OP-based Evasion Attack	HP-based Poisoning Attack	OP-based Poisoning Attack
Underlying Reasons of Vulnerability	✓	✓	○	○
Effectiveness	✓	✓	○	✓
Efficiency	○	✓	○	✓
Imperceptibility	✓	✓	○	✓
Comparison between different models	✓	✓	○	✓
Transferability	○	✓	○	✓
Attack against Defense	○	✓	○	✓
Limitations of the Optimization Objective	—	✓	—	○
Diverse Forms	○	✓	○	✓
Perturbation Constraints	○	✓	○	○
Supervision-agnostic Attack	—	✓	—	✓
Against diverse models	✓ (CNNs, ViTs, MLLMs)	✓ (CNNs, ViTs, MLLMs)	○	✓ (CNNs, ViTs)

TABLE IV
CHALLENGE-ORIENTED COMPARISON OF REPRESENTATIVE ROBUSTNESS STRATEGIES. ✓: EXPLORED; ○: NOT EXPLICITLY STUDIED; —: NOT APPLICABLE.

Challenge Dimension	Augmentation	Input Pre-processing	Randomized Smoothing	Adversarial Training
Certified robustness	○	○	✓	○
Overfitting	✓	✓	✓	✓
Efficiency	○	✓	✓	✓
Robust fairness	○	○	○	✓
Robustness-accuracy trade-off	○	○	✓	✓
Defense against patch attack	○	○	○	✓
Multi-attack robustness	○	✓	✓	✓
Pre-training and fine-tuning robustness	○	○	○	✓
Adaptive learning	○	○	○	✓
Against evasion attack	✓	✓	✓	✓
Against poisoning attack	✓	✓	✓	✓

robustness, certified robustness and so on. We also suggest directions for future research, including complex scenarios involving the co-existence of multiple perturbations, multimodal attacks and defenses, the exploration of more advanced models, and unified perturbations. Through this survey, we aim to provide a structured overview of the landscape and reveal critical properties of deep models, thereby contributing to the development of trustworthy real-world systems (more content can be seen in the supplementary material).

A. Distinctions From Existing Surveys

Some earlier surveys were published several years ago and therefore cannot adequately reflect the rapid recent developments in image perturbation research [39], [40], [41], [54]. More importantly, many existing works lack comprehensive exploration of diverse perturbation types [39], [41], intrinsic vulnerability mechanisms under perturbations [39], the relationships among different attack paradigms [39], [40], [41],

[55], [56], [57], and the connections between perturbation-based attacks and defenses [39], [40], [41], [42], [55], [56], [57]. In addition, prior surveys provide limited discussion of critical challenges, and solutions in this field [39], [40], [41], [42], [43], [54], [55], [56], [57].

In comparison, our distinct contributions are as follows:

1) *Perturbation-oriented perspective*: Unlike conventional surveys that mainly organize the literature according to attack formulations or application scenarios, our survey adopts a perturbation-oriented perspective that unifies attacks and defenses through the lens of perturbation characteristics and their impacts on model robustness.

2) *Comprehensive coverage and clear taxonomy*: Our survey provides comprehensive coverage together with a clear and systematic taxonomy of perturbation-driven attacks, defenses, challenges and solutions across diverse learning paradigms.

3) *Comprehensive intrinsic vulnerability analysis*: Beyond summarizing attack methodologies, our survey emphasizes intrinsic vulnerability mechanisms under perturbations and further

analyzes their behaviors across different model architectures. This perspective helps reveal shared robustness weaknesses, improves the understanding of model vulnerability evolution, and may further inspire the design of more robust future architectures.

4) *Intrinsic interconnections among different paradigms:* Our survey investigates the shared perturbation principles and intrinsic interconnections among diverse attack and defense paradigms, rather than reviewing them independently. This unified perspective helps explain how perturbations fundamentally influence model behaviors and robustness across various settings, including their dual role in both exposing model vulnerabilities through attacks and improving robustness through defenses. Such a perspective provides a more principled understanding of robustness learning and the co-evolution of attacks and defenses, and may further inspire the development of more generalized and effective robustness strategies.

II. DEFINITION, TAXONOMY AND APPLICATIONS

A. Definition and Taxonomy

Data perturbations are modifications to data that occur during data collection or processing before being input into models. Visual data from the real world often contain perturbations due to factors like transmission, compression, sensor errors, environmental conditions (e.g., fog, snow, lighting), and deliberate attacks. Additionally, data may undergo further adjustments, such as adding noise or applying transformations, either to facilitate adversarial attacks or enhance model robustness. A classification of perturbations is shown in Table I with an exemplary visualization in Fig. 1.

Depending on whether optimization is involved in the *generation process*, data perturbations can be categorized into *heuristic perturbations (HP)* and *optimization-guided perturbations (OP)*. HP-based methods apply predefined, input-agnostic transformations or rules, without involving an explicit optimization procedure. OP-based methods, in contrast, obtain perturbations by solving a well-defined optimization problem with respect to a specified objective. Furthermore, both types can be classified as non-adversarial or adversarial perturbations based on the *presence of a specific attack intention*. Non-adversarial perturbations modify data without a specific attack goal, whereas adversarial perturbations are deliberately crafted for a specific target.

From the perspective of *magnitude*, current perturbations can be classified as bounded and unbound categories, depending on whether a magnitude constraint is applied. Heuristic and non-adversarial perturbations are generally not subject to strict magnitude constraints. Optimization-guided and adversarial ones are often bounded to maintain imperceptibility (in attacks) or model performance (in robust learning). However, in real-world applications, adversarial perturbations are often unbounded to increase attack success rates [58], [59], [60]. Various distance metrics are used, including l_p , $p \in \{0, 1, 2, \infty\}$ distance, CIEDE2000 [61], [62], SSIM [62], [63], LPIPS [62], [64], Wasserstein distance [65], geodesic distance [66], and parameter size based distance [67].

From the *scale*, perturbations can be single-pixel [68], patch-level [58], [69], [70], whole-area [14], [71], [72], and distribution-level [73], affecting individual pixels, localized parts, entire samples, and the overall data distribution, respectively. Non-adversarial perturbations often target entire images while adversarial ones encompass all these types.

From the *scope*, perturbations can target specific instances (i.e., sample-wise perturbations [1], [2], [74], [75]), specific classes (i.e., class-wise perturbations [76], [77]), and all data (i.e., universal perturbations [78], [79]).

According to the availability of *knowledge*, optimization-guided perturbation generation can be categorized into white-box, black-box, and no-box scenarios. In the white-box scenario, adversaries have full access to target models, training algorithms, and training data. In the black-box case, adversaries have only access to the model output or data. In the no-box case, adversaries have no access to any information about models and data. Heuristic perturbations fall under the no-box category, as they are independent of the victim model.

B. Applications

As the saying goes, “water can carry a boat, but can also overturn it”, perturbations can be crafted to challenge model robustness at both training and testing. Conversely, they can also be used to enhance model robustness. (Table II and Fig. 2).

1) *Perturbation-Based Attack: Evasion attack* applies adversarial perturbations to input data during the testing or deployment phase to degrade model performance. Formally, denote the original image as x_{test} . The objective of an evasion attack is to seek perturbation operation $\xi(\cdot)$ on the original image that results in an adversarial example $x_{\text{adv}} = \xi(x_{\text{test}})$ that satisfies:

$$\begin{aligned} \text{Untargeted Attack: } & \max_{\xi} (f_{\theta}(\xi(x_{\text{test}})) \neq p_{x_{\text{test}}}), \\ \text{Targeted Attack: } & \min_{\xi} (f_{\theta}(\xi(x_{\text{test}})) = y_t), \end{aligned} \quad (1)$$

where y_t is the target output. $M(\xi(x_{\text{test}}), x_{\text{test}}) \leq \epsilon$ are usually applied to ensure that perturbations remain imperceptible, e.g., $\|\xi(x_{\text{test}}) - x_{\text{test}}\|_p \leq \epsilon$. $p_{x_{\text{test}}}$ can be the ground truth of x_{test} , i.e., $p_{x_{\text{test}}} = y_{\text{test}}$, or the model's output for x_{test} . Perturbations can take various forms, such as additive noise represented by $\xi(x_{\text{test}}) = x_{\text{test}} + \delta$, where δ is the noise.

Poisoning attack corrupts the training process of models by injecting perturbations into part or all of the training data, aiming to manipulate models' behaviour at test time:

$$\begin{aligned} \text{Untargeted Attacks: } & \max_{\xi} (f_{\hat{\theta}}(x_{\text{test}}) \neq p_{x_{\text{test}}}), \\ \text{Targeted Attacks: } & \min_{\xi} (f_{\hat{\theta}}(x_{\text{test}}) = y_t), \\ \text{Backdoor Attacks: } & \min_{\xi} (f_{\hat{\theta}}(x_{\text{test}} + x_{\text{trig}}) = y_t), \\ \text{s.t. } & \hat{\theta} = \arg \min_{\theta} \mathcal{L}(f_{\theta}, \hat{\mathcal{D}}_{\text{train}}), \end{aligned} \quad (2)$$

where $\mathcal{D}_{\text{clean1}}$ and $\mathcal{D}_{\text{clean2}}$ contain clean samples. $\mathcal{L}$ is the training loss for the model. $\hat{\mathcal{D}}_{\text{train}} = \mathcal{D}_{\text{clean1}} \cup \mathcal{D}_{\text{adv}}$, $\mathcal{D}_{\text{adv}} =$

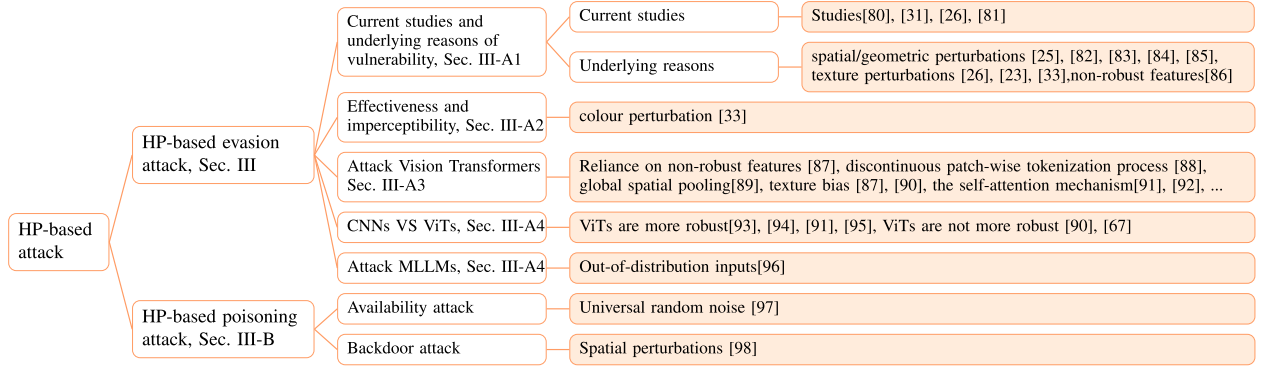


Fig. 3. A structured overview of HP-based attack covering generation methods, key challenges, and solution strategies. Further details are provided in the main text.

$\{(\xi(x_{\text{clean2}}), y_{\text{adv}})\}_{i=1}^m, (x_{\text{clean2}}, y_{\text{clean2}}) \in \mathcal{D}_{\text{clean2}}, M(\xi(x_{\text{clean2}}), x_{\text{clean2}}) \leq \epsilon, y_{\text{adv}}$ can either be the true label, i.e., $y_{\text{adv}} = y_{\text{clean2}}$, or another label.

2) *Perturbation-Based Defense*: While perturbations are used in adversarial attacks to expose model vulnerabilities, this same property enables their use in defense, where incorporating them during training or inference helps mitigate these weaknesses and improve robustness.

Perturbations are used to enhance model robustness against attacks through various methods, including augmentation (Section III-C), input preprocessing (Section III-D), randomized smoothing (Section III-E), and adversarial training (Section IV-D). As these methods apply perturbations in different ways, no universal definition exists. We will introduce each method separately.

Most attack and defense methods concentrate on convolutional neural networks (CNNs). Unless stated otherwise, methods discussed will pertain to them, with separate sections dedicated to others, like vision transformers (ViTs) and large language models (LLMs).

C. Rationale for Categorizing Perturbations

Before introducing heuristic and optimization-guided perturbations in detail, we first clarify and emphasize the rationale behind this categorization. From a theoretical perspective, optimization-guided perturbations explicitly maximize a loss function to identify worst-case directions, whereas heuristic perturbations (e.g., random or rule-based transformations) sample from a transformation space without explicitly optimizing an objective. This indicates different search behaviors. Optimization-based methods perform local, model-driven exploitation while heuristic methods provide diverse, model-agnostic exploration, sampling multiple directions that gradients may not reach due to locality, non-differentiability, or objective bias. As a result, heuristic perturbations can uncover diverse and realistic failure modes (e.g., occlusion, compression and style alteration [26], [31], [80]), while optimization-based methods quantify the worst-case vulnerability [2], [20], [99]. In other words, these perspectives are complementary, providing coverage and attack strength, respectively.

In defenses, optimization-based methods (e.g., adversarial training) enforce robustness under worst-case perturbations, but are typically limited to specific perturbation models and may not generalize well to broader distributional variations. In contrast, heuristic methods (e.g., randomized smoothing) enhance robustness to diverse and stochastic input variations, improving generalization under distributional shifts, but do not explicitly target worst-case adversarial directions. As a result, they are typically less effective against strong adversarial attacks. Achieving stronger robustness would require broader coverage of the perturbation space, which may introduce additional computational overhead. To this end, recent studies have begun to explore combining these approaches to achieve more principled robustness [100], [101], [102], [103]

III. HEURISTIC PERTURBATION (HP)

A. HP-Based Evasion Attack

An overview of HP-based attacks and defenses is provided in Figs. 3 and 4.

1) *Current Studies and Underlying Reasons of Vulnerability*: The vulnerability of CNNs to heuristic perturbations has been extensively studied. Research indicates that simple perturbations can significantly destabilize CNNs. For instance, Dodge et al. [26], [31], [80] develop benchmarks to demonstrate the negative impact of different perturbations, including blur, additive noise, grayscale conversion, weather effects, style alterations, and so on. Vasiljevic et al. [81] highlight the weakness to optical blur. The weakness of CNNs to spatial/geometric perturbations has also been widely investigated [25], [67]. A key factor in spatial weaknesses is the subsampling operations in CNNs, such as strided-convolution, max-pooling, and average pooling, which ignores the classic sampling theorem and causes aliasing [25], [82], [83], [84], [85] (Aliasing refers to the phenomenon that high-frequency signals degenerate into completely different ones after sampling). Under aliasing effects, spatial/geometric perturbations can significantly change high-frequency signals, resulting in different samples. Additionally, texture perturbations have been shown to mislead CNNs due to CNNs' strong reliance on texture information [23], [26], [33]. Benz et al. [86] claim that batch normalization and other normalization variants

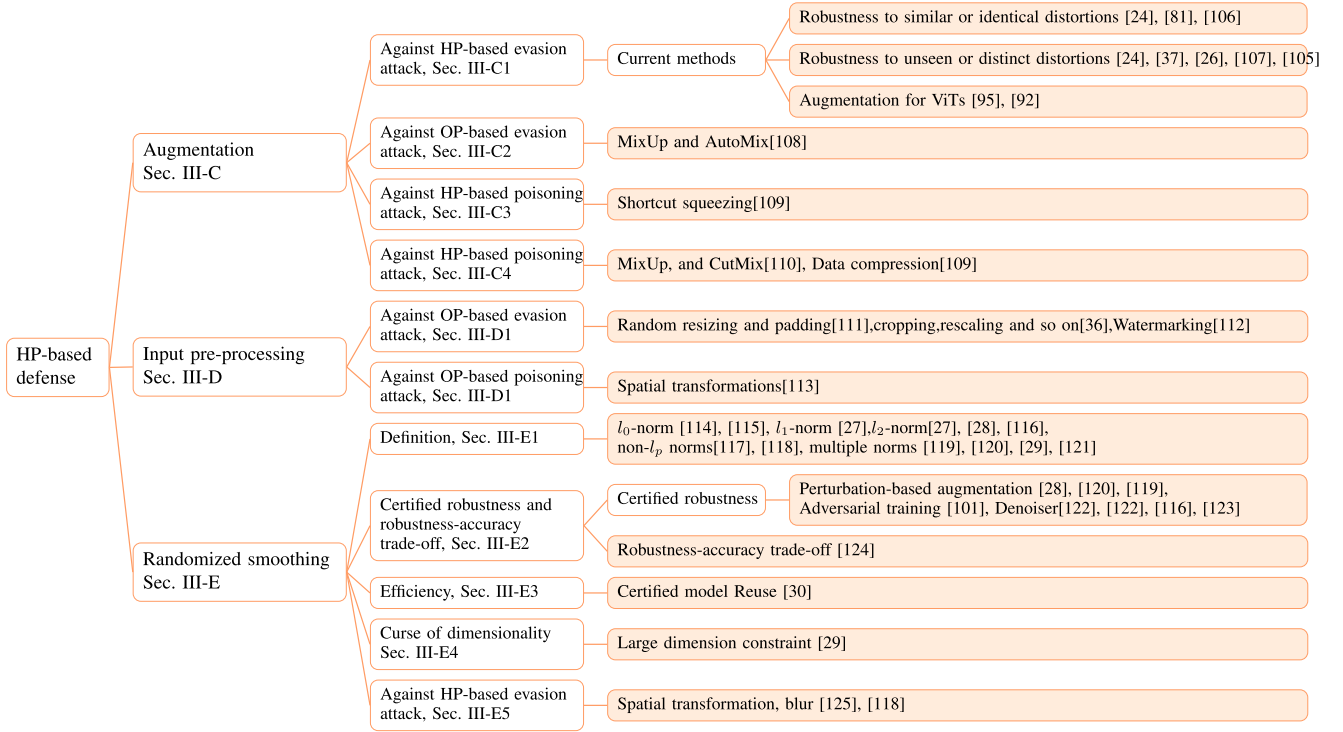


Fig. 4. A structured overview of HP-based defense covering generation methods, key challenges, and solution strategies. Further details are provided in the main text.

increase vulnerability as they shift the model to rely more on non-robust features than robust features.

2) *Effectiveness and Imperceptibility*: The effectiveness of heuristic perturbations is limited by their heuristic generation process. To handle this, some propose optimization-guided perturbations [66], [67], [85], [104]. We will introduce it in Section IV-A. In addition, some approaches learn perturbations by modifying colour, making them appear more natural [33]. However, without additional constraints, these perturbations are still perceptible to humans.

3) *Attack Vision Transformers*: Studies show that similar to CNNs, ViTs are sensitive to both heuristic and optimization-guided perturbations [92], [94], [95], [105]. It is related to several factors, including reliance on non-robust features [87], the discontinuous patch-wise tokenization process [88], global spatial pooling (for spatial perturbations) [89], texture bias (for style perturbation) [87], [90], the self-attention mechanism (for patch-wise perturbations) [91], [92]. The robustness of ViTs is also influenced by patch size [90], [91], [94], model size [67], [90], [94], dataset size [94], object class [94], training procedure [90], [94], and architectural structure [91].

4) *Comparison Between CNNs and ViTs*: ViTs and CNNs share some similarities and differences when facing perturbations. Naseer et al. [91], [93], [94], [95] insist that ViTs are generally more robust to heuristic perturbations, while others disagree [67], [90]. Specifically, Bhojanapalli et al. [90] show that ViTs are less robust than ResNets (a kind of CNNs) when trained on small datasets, and simply increasing the model size of ViTs does not guarantee better robustness. However, with larger datasets, increasing model size or reducing patch size

can improve the robustness of ViTs, outperforming ResNets. Larger models enhance robustness to perturbations for both, with the effect more pronounced in ViTs. Notably, ViTs are particularly sensitive to patch noise due to their self-attention mechanism [92].

5) *Attack Multimodal Large Language Models (MLLMs)*: MLLMs have received considerable attention due to powerful multimodal generation. Some works began to explore their vulnerabilities. For example, [96] create out-of-distribution inputs through data transformations for attacks.

B. HP-Based Poisoning Attack

Studies of heuristic perturbations for poisoning attacks are still in the early stages, with few works proposed [97], [98]. For availability attacks, Fowl et al. [97] show that universal random noise on training samples can deter models from learning knowledge. For backdoor attacks, Xu et al. [98] apply spatial perturbations (e.g., rotation) with the specific parameter (e.g., angle) into training data and assign a target label. Images rotated to a particular angle (as triggers) will activate the backdoor in models trained with poisoned images. The reasons for this vulnerability remain unexplored. A comparison of different attack strategies is provided in Table III.

C. Augmentation for Defense Via HP

It increases training data diversity by applying transformations to improve the invariant learning ability of models.

1) *Defend Against HP-Based Evasion Attack. Current Methods:* Heuristic perturbations for augmentation typically improve robustness to similar or identical distortions, e.g., random erasing to occlusion [24], blur to blur [81], geometric augmentations to motion blur [106]. However, they often fail to ensure invariance to unseen or perceptually distinct perturbations [25], [67], [106], [126]. Enhancing robustness against a wider range of perturbations requires training models with diverse perturbation types [24], [26], [37], [105], [107]. For example, Yin et al. [107] utilize AutoAugment [127] to enhance robustness against diverse perturbations. Hendrycks et al. [37] randomly sample perturbation operations from AutoAugment to construct augmentation chains or compositions and use the mix operation to produce diverse data. Hendrycks et al. [26] randomly sample manually designed functions, e.g., zeroing, convolving, and applying activation functions to generate semantically consistent images with diverse perturbations against blur. Müller et al. [105] augment data with optical kernels and leverage AugMix to improve robustness to optical blur and other perturbations. Notably, due to the characteristics of CNNs, certain perturbations can enhance robustness against a range of perturbations. For instance, Geirhos et al. [23] show that CNNs are biased towards textures over shapes. Leveraging style perturbations to promote shape bias can enhance model robustness to various perturbations, including uniform noise, contrast, high/low-pass filtering, and phase noise.

Augmentation for ViTs: Given the patch-wise data processing of ViTs, defense often targets the patch level. For instance, Mao et al. [95] propose patch-wise augmentation to improve robustness to heuristic and optimization-guided adversarial perturbations. They indicate that robustness improvement mainly stems from the strategy itself rather than the used augmentation. Guo et al. [92] identify and occlude the most vulnerable patches and reduce their sensitivity to perturbations by aligning the intermediate features between clean and corrupted examples.

2) *Defense Against OP-Based Evasion Attack:* Techniques such as MixUp and AutoMix can only offer slight robustness to single-step adversaries, i.e., FGSM [108].

3) *Defense Against HP-Based Poisoning Attack:* Liu et al. [109] propose image shortcut squeezing, which leverages data compression techniques to alleviate the negative influence of data poisoning.

4) *Defense Against OP-Based Poisoning Attack:* Borgnia et al. [110] evaluate the effectiveness of augmentation, e.g., MixUp, and CutMix, for defending against simple backdoor and targeted poisoning attacks. It is found that for backdoor attacks, CutMix is more effective as the cut operation is more likely to break trigger patterns while MixUp cannot. For targeted poisoning attacks, MixUp show advantages, and the reason might be that it would incur the implicit linearity between data points. In adaptive attack settings, CutMix is more effective because attackers are unaware of the cutting locations. Liu et al. [109] leverage data compression techniques to eliminate the negative influence of data poisoning. However, heuristic perturbation-based augmentation does not always work towards

eliminating perturbations. For example, Yu et al. [77] implement availability attacks by adopting a padding operation to introduce local correlation into the perturbations, thus making them robust to strong augmentation, such as Cutout, Mixup, CutMix, and Fast AutoAugment.

D. Input Pre-Processing for Defense Via HP

This approach employs specific perturbation operations as a preprocessing step to disrupt the perturbation patterns.

1) *Defense Against OP-Based Evasion Attack:* Xie et al. [111] show random resizing and padding can destroy optimization-guided perturbations, including FGSM [2], DeepFool [75] and C&W [74]. Guo et al. [36] find that perturbations based on image cropping and rescaling, bit-depth reduction, JPEG compression, total variance minimization and image quilting can defend single-step and iterative attacks. But they would fail when facing strong attacks [128]. Hayes et al. [112] leverage watermarking to block adversarial patches by measuring saliency maps.

2) *Defense Against OP-Based Poisoning Attack:* Li et al. [113] demonstrate that spatial transformations, such as flipping and shrinking, can disrupt static backdoor triggers with fixed locations and appearances.

E. Randomized Smoothing (RS) for Defense

A comparison of different defense strategies is provided in Table IV.

1) *Definition:* RS aims to ensure *certified robustness* against evasion attacks, which means that no future adversary (under a certain threat model) will break the defense. Specifically, RS transforms a base classifier f_θ into a “smoothed” classifier g_ϕ by injecting random noise into the input data and averaging predictions over multiple noisy samples. The added noise serves as a form of regularization, smoothing the model’s decision boundaries and making it less vulnerable to adversarial perturbations. The smoothed classifier g_ϕ predicts the label of a query x as the one that is most likely to be returned by the base classifier under random noise on x :

$$g_\phi(x) = \arg \max_{c \in \mathcal{Y}} \mathbb{P}(f_\theta(x + \varrho) = c), \quad (3)$$

where ϱ denotes noise, e.g., Gaussian noise $\varrho \sim \mathcal{N}(0, \sigma^2 I)$. σ is a hyperparameter to balance robustness and accuracy. $\mathbb{P}(f_\theta(x + \varrho) = c)$ is the probability that the base classifier predicts class c after noise ϱ is added. Let p_A and p_B represent the probabilities of classes from the base classifier for perturbed samples, respectively. The robustness of the smoothed classifier is guaranteed within the l_2 ball around x with radius $r = \frac{\sigma}{2}(\Phi^{-1}(p_A) - \Phi^{-1}(p_B))$ [27], [28], where Φ^{-1} is the inverse of the standard Gaussian CDF. In other words, g_ϕ is robust iff $g_\phi(\xi(x)) = g_\phi(x) \forall \|\varrho\|_2 \leq r$.

Beyond l_2 -norm Robustness Certification: Conventional RS primarily focuses on l_2 -norm robustness [28], [116]. Many works extend this robustness to other types, such as l_0 [114], [115], l_1 [27], and non- l_p norms (e.g., Wasserstein adversarial perturbations [117], semantic transformations-based adversarial

perturbations [118]). Additionally, robustness has been explored for heuristic adversarial semantic perturbations [118], [125]. Other works address multiple norms simultaneously, such as l_1 , l_2 [119], [120], and l_1 , l_2 , l_∞ [29], [121].

Beyond Gaussian Noise: In addition to Gaussian noise [27], [28], [118], [119], various perturbations have also been explored, including Laplacian noise [27], [117], [118], [120], uniform noise [115], [118], [119], noise from discrete distribution [115], semantic transformation [125], and random ablations [114].

2) Certified Robustness and Robustness-Accuracy Trade-Off: Using Gaussian or other perturbations to assist predictions, standard RS requires pre-trained models inherently robust to them [28]. However, this robustness is not always feasible [122]. To address this, some propose to train a more robust base classifier through perturbation-based augmentation [28], [119], [120], adversarial training [101], and direct certified radius maximization [129]. Pal et al. [130] claim that augmentation might not always be effective. Instead of enhancing the base classifier, some propose adding a denoiser as a preprocessor to remove noise (i.e., denoised smoothing), such as DnCNN [122], MemNet [122], and diffusion models [116], [123]. Some attackers may only disturb a subset of all entities, e.g., pixels in images. However, existing RS methods typically add noise into the entire input, which can degrade standard accuracy. To handle this, Scholten et al. [124] propose to perturb a subset of data entities.

3) Efficiency: Randomized smoothing requires inferring a large number of corruptions per input, which is time-consuming. Ugare et al. [30] reuse information from a certified model to accelerate the certification of similar models.

4) Curse of Dimensionality: RS with Gaussian distribution is effective against l_2 adversarial perturbations but loses efficacy when dealing with $l_{p>2}$ -norm adversaries in the high-dimensional regime. Specifically, the largest certified radius r decreases with the number of dimension d as $O(d^{\frac{1}{p}-\frac{1}{2}})$. In other words, when the input dimension d is large, they cannot achieve certified nontrivial accuracy at more than $\Omega(\min(1, d^{\frac{1}{p}-\frac{1}{2}}))\ell_p$ -radius. To handle this, it is recommended to consider more information about the base classifier or the multi-class problem structure [29].

5) Defend Against HP-Based Evasion Attack: Most RS methods concentrate on defending against optimization-guided adversarial perturbations. Some began to extend RS to deal with heuristic adversarial perturbations, including spatial transformation, blur and so on [118], [125].

IV. OPTIMIZATION-GUIDED PERTURBATION (OP)

While heuristic perturbations provide broad, model-agnostic exploration of the perturbation space, they may fail to identify the most vulnerable directions of a model, leading to an underestimation of its vulnerability. To address this limitation, optimization-guided perturbations are proposed to uncover worst-case vulnerabilities and promote robustness. An overview of HP-based attacks and defenses is provided in Figs. 5, 6, and 7.

A. OP-Based Evasion Attack

1) Basic Generation Methods: L-BFGS. Szegedy et al. [11] first demonstrate that properly designed slight perturbations would mislead DNNs to make wrong predictions. L-BFGS is proposed to learn perturbations by solving a box-constrained optimization problem:

$$\min_{\delta} \|\delta\|_2, \text{ s.t. } f_{\theta}(x + \delta) = y_t, \quad x + \delta \in [0, 1]^d, \quad (4)$$

where $y_t \neq y$ is the target label. A box-constrained L-BFGS is used to obtain an approximate solution by line search.

C&W. Carlini et al. [74] propose to bypass directly solving the constraint $f_{\theta}(x + \delta) = y_t$ in Eq.(4).

DeepFool. Moosavi et al. [75] show that the robustness of an affine classifier $f = w^T x + b$ at x can be actually calculated as the distance from x to the separating affine hyperplane $\mathbf{F} = \{x : w^T x + b = 0\}$. The perturbation is iteratively adjusted until it alters the classifier's output sign, indicating successful deception of the classifier.

FGSM. Goodfellow et al. [2] exploit the "linearity" of DNNs and propose a one-step Fast Gradient Sign Method (FGSM) to perturb the input data in the gradient direction that increases the loss: $x_{\text{adv}} = \Pi_{x,\epsilon}\{x + \epsilon \cdot \text{sign}(\nabla_x \ell(f_{\theta}(x), y))\}$, where $\Pi_{x,\epsilon}$ is a clip function to constrain perturbed images within the range of ϵ -neighbourhood of original images.

Iterative FGSM utilizes a small step size to increase success rates [99], [131]: $x_{\text{adv}}^{i+1} = \Pi_{x,\epsilon}\{x_{\text{adv}}^i + \pi \cdot \text{sign}(\nabla_x \ell(f_{\theta}(x_{\text{adv}}^i), y))\}$, where i is the iterative step. $\pi = \epsilon/T$ is the step size with T as the iterative time.

Projected Gradient Descent (PGD) [20] is another iterative variant of FGSM. The difference between PGD and Iterative FGSM is that PGD utilizes a random start in the allowed norm ball around the original example.

Momentum Iterative FGSM leverages momentum to stabilize update directions and help escape poor local maxima [188]: $x_{\text{adv}}^{i+1} = \Pi_{x,\epsilon}\{x_{\text{adv}}^i + \pi \cdot \text{sign}(g^{i+1})\}$, s.t. $g^{i+1} = \mu \cdot g^i + \frac{\nabla_x \ell(f_{\theta}(x_{\text{adv}}^i), y)}{\|\nabla_x \ell(f_{\theta}(x_{\text{adv}}^i), y)\|_1}$, where μ is the decay factor.

Jacobian-based Saliency Map methods generate adversarial perturbations based on forward gradient propagation. For instance, Papernot et al. [71] measure the influence of input changes on outputs through the forward derivative, interpreted as the Jacobian matrix of the loss function.

Generative methods leverage generative models to produce perturbations, including residual-like networks [193], encoder-decoder networks [132], [133], [134], [193], and generative adversarial networks [194].

Search-based methods find optimal solutions by systematically exploring a space of possible inputs or configurations until it successfully misleads target models, including random search methods, exhaustive search methods and population-based methods. **Random search methods** produce perturbations by sampling random combinations of parameters or configurations, e.g., hue and saturation components [33], and different channels [34]. **Exhaustive search methods** explore all possible perturbations within a defined space [60], [67], [85]. **Population-based methods** maintain a population of candidate

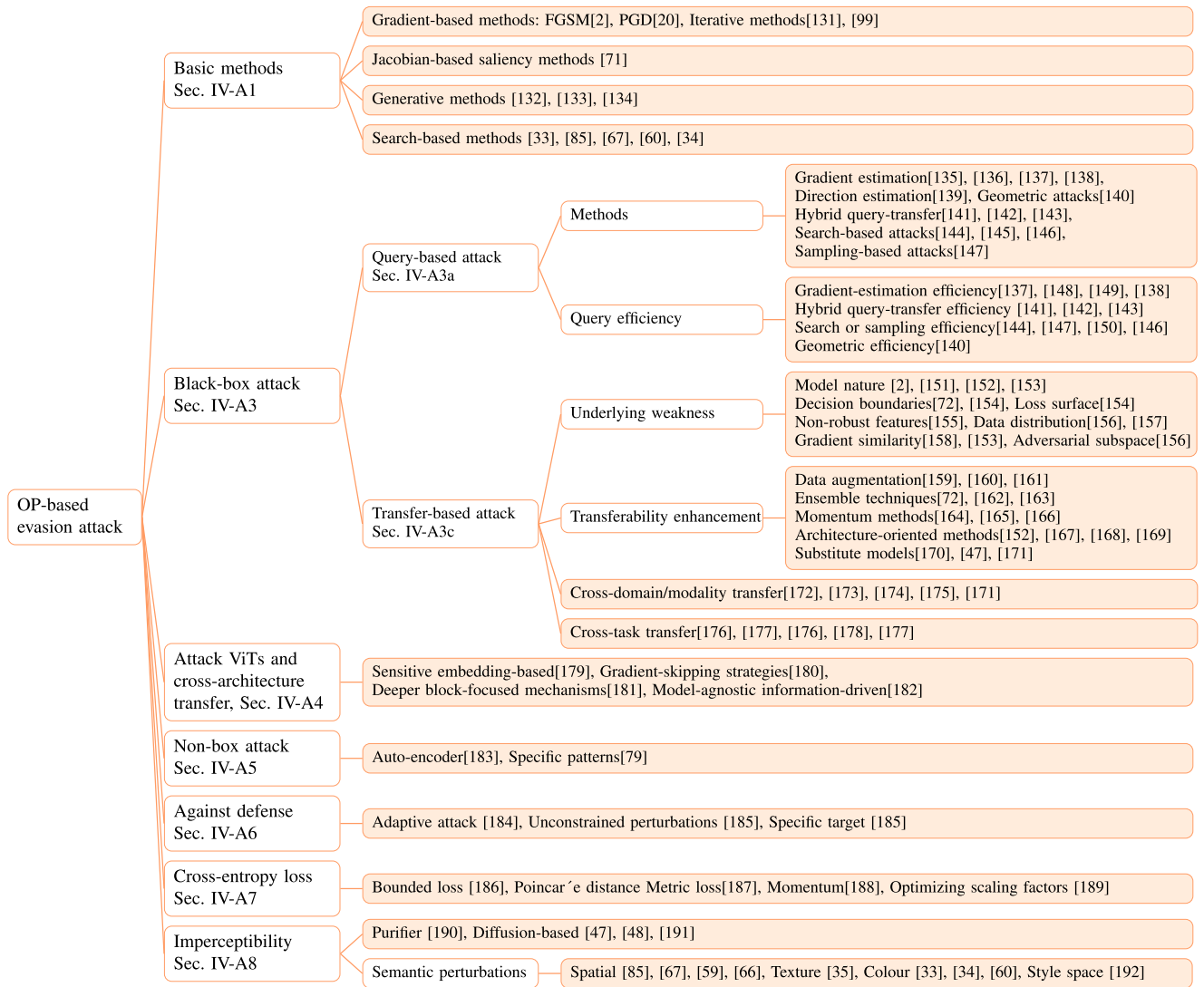


Fig. 5. A structured overview of OP-based evasion attacks covering generation methods, key challenges, and solution strategies. Further details are provided in the main text.

solutions, which are iteratively updated through evolutionary processes [68], [195].

2) *Underlying Reasons of Vulnerability*: The adversarial vulnerability of CNNs is often attributed to their linear nature [2] and reliance on non-robust features [155], [196] or spurious cues (shortcuts) [197], [198], [199]. Goodfellow et al. [2] state that DNNs behave in a linear way, where data is processed as a way of high-dimensional dot products between weights and data. This implies that the change in activation resulting from perturbation can increase linearly with the dimensionality of the weight vectors. In this context, minor perturbations can accumulate to produce a significant change in the output. Non-robust features or spurious cues (shortcuts) refer to features that fail to capture the true semantic essence of objects in images and are unstable across different conditions. These include the background context [198], co-occurrence cues [197], textures [23], styles [199] and noise [155]. Adversarial perturbations can alter them, disrupting the link between them and the target [197],

[199]. Some claim that the reliance on non-robust features is partly attributed to the use of batch normalization (BN) and other normalization variants [86]. Other reasons can refer to heuristic adversarial perturbations in Section III-A-1. Reasons for ViTs will be introduced in Section IV-A-4.

3) *Black-Box Attack*: In white-box settings, adversaries have full access to the victim model, training algorithm and data, making it easy to mount effective attacks. In contrast, in black-box scenarios, limited knowledge of these details hinders learning effective perturbations. To address this, current solutions include *query-based attacks* [14], [136], [138], [200] and *transfer-based attacks* [72], [162], [164], [201]. Query-based attacks iteratively query the target model to craft adversarial perturbations based on the model's feedback. Transfer-based methods use substitute models as the attacking target. Both methods rely on adversarial transferability to some extent, meaning that adversarial examples generated to attack one model may also be effective against other models [2], [20], [156]. Transfer-based methods

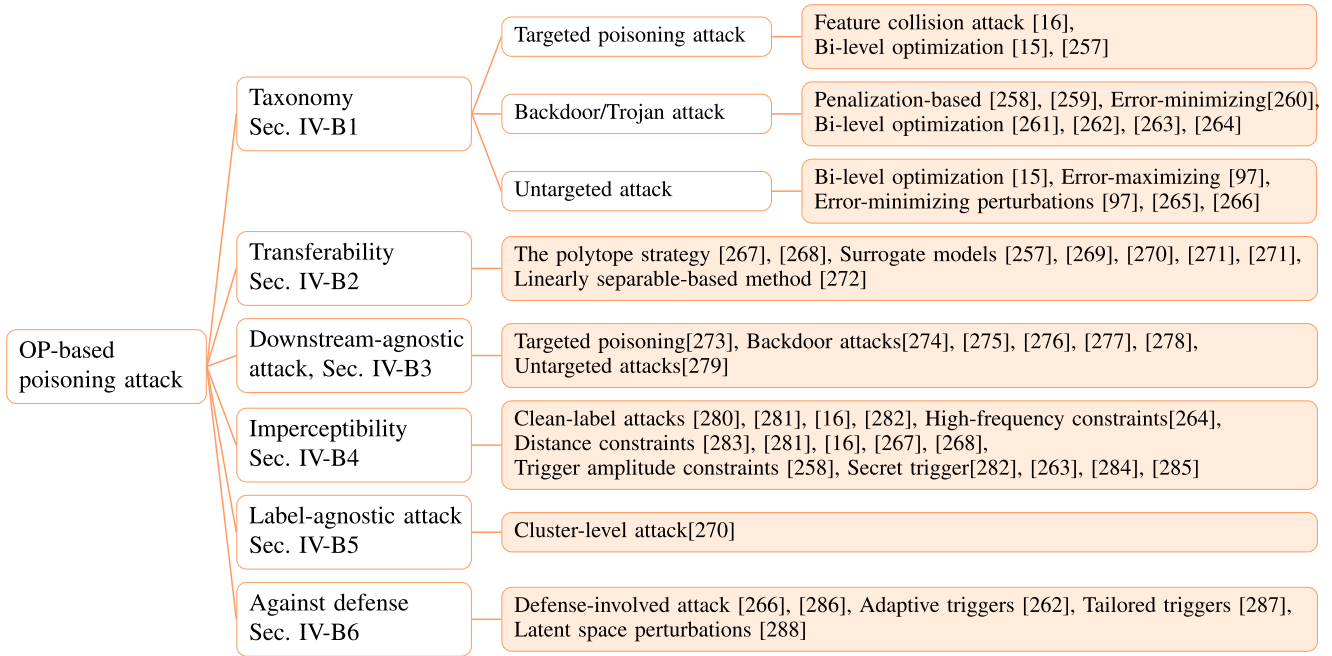


Fig. 6. A structured overview of OP-based poisoning attacks covering generation methods, key challenges, and solution strategies. Further details are provided in the main text.

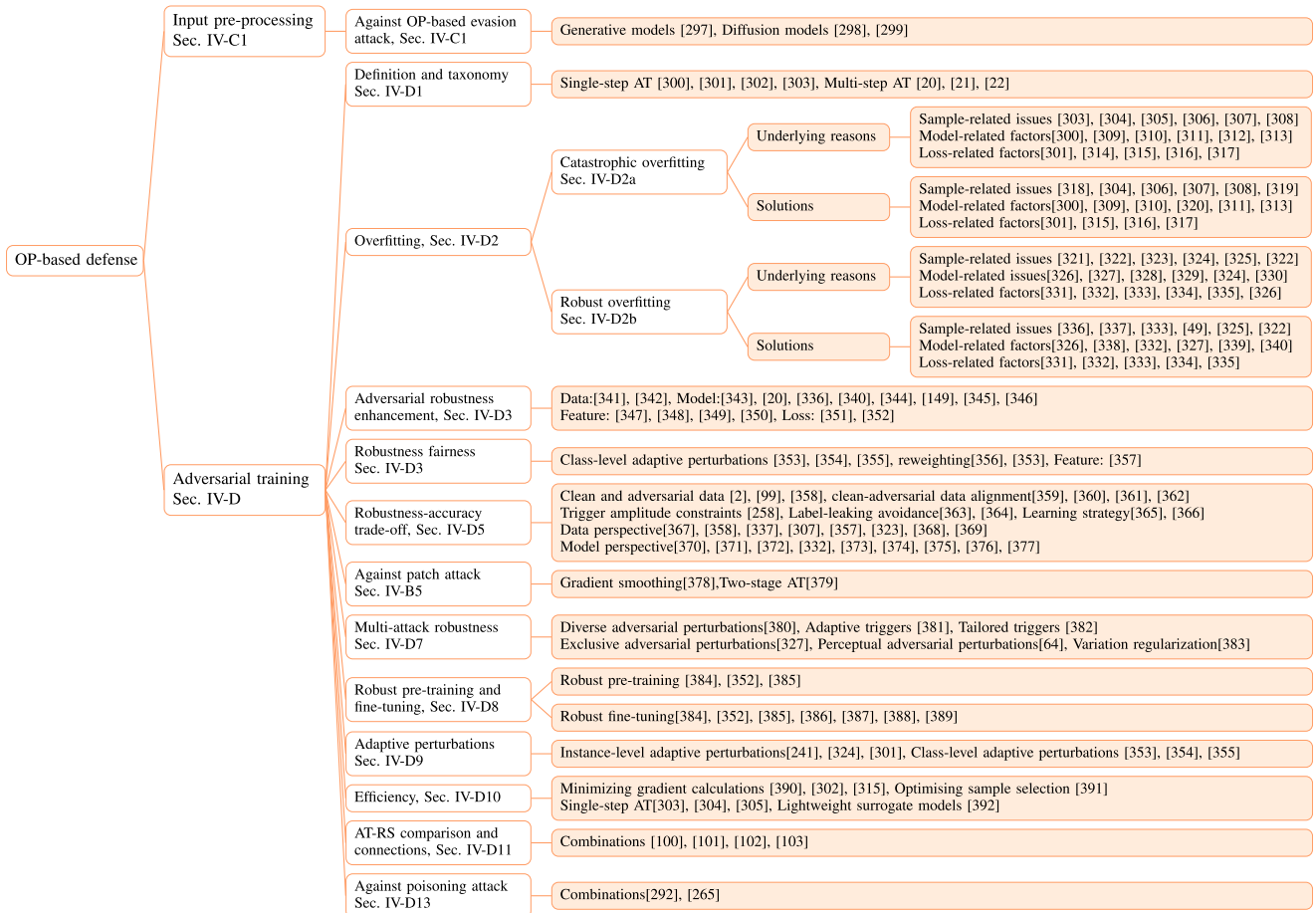


Fig. 7. A structured overview of adversarial training covering generation methods, key challenges, and solution strategies. Further details are provided in the main text.

are more flexible as they don't require target model feedback but are generally less effective than query-based methods. Often, the two approaches are combined for better performance [133], [141], [142], [143].

a) Query-based Attack: In black-box settings, the lack of direct access to the internal configurations of models makes it hard for the gradient-based method to generate perturbations. To handle this, many query-based attacks use derivative-free optimization methods, including gradient estimation based attacks [133], [134], [135], [136], [137], [138], [202], [203], gradient direction estimation based attacks [139], the combination of query-based and transfer-based attacks [141], [142], [143], search-based attacks [144], [145], [146], sampling-based attacks [147] and geometric-based attacks [140].

b) Query Efficiency: Query-based methods require numerous queries to achieve high success rates, resulting in significant computational costs. For instance, Chen et al. [135] utilize the finite difference method to estimate gradients, but its query complexity grows with data dimensionality due to the coordinate-wise gradient estimation. Similarly, searching or sampling-based methods also need numerous queries to find optimal perturbations [144], [147]. Moreover, in real-world scenarios, the limited availability of data makes such extensive querying impractical.

To reduce queries, various methods have been proposed. For gradient-estimation-based approaches, Bhagoji et al. [136] estimate gradients for groups of features rather than individual features. Instead of finite difference [135], [136], Ilyas et al. [137] use a least squares method to avoid precise estimation of the gradient, and taking advantage of the prior of the gradient. Ilyas et al. [202] leverage the Natural Evolution strategy based on the search distribution. Additional improvements include leveraging the prior of pre-trained source models [133], modelling adversarial distributions [149], [203] and utilizing the structure information of images [149]. Tu et al. [134] propose a scaled random full gradient estimation and use an autoencoder to reduce the attack dimension. Previous methods [133], [134], [134], [135], [136], [137], [149], [203] that utilize zeroth-order gradient estimation would adopt the first-order gradient descent methods to produce perturbations. Zhao et al. [138] instead leverage the second-order natural gradient descent to accelerate query. Chen et al. [139] employ Monte Carlo approximation to estimate the gradient direction. Li et al. [204] focus on searching for a small, representative subspace for query generation. Shen et al. [148] utilize substitute models with similar functions to the target model, effectively reducing the number of queries needed to approximate the target model.

Several approaches combine transfer-based and query-based attacks to accelerate query. For instance, Cheng et al. [141] use transfer-based prior from the gradient of a surrogate model. Cai et al. [142] refine the surrogate ensemble by querying the victim model to reduce the search space dimension. Ma et al. [143] train a generalized substitute model via meta-learning, enabling rapid adaptation to target models.

To improve the efficiency of search-based and sampling-based methods, Croce et al. [144], [145] employ specific sampling

distributions to facilitate the search for possible solutions. Brunner et al. [147] incorporate priors such as low-frequency, regional masking, and gradients from surrogate models. Sarkar et al. [150] apply reinforcement learning to develop an optimum policy for adversarial attacks, functioning like a deep tree search rather than an exhaustive search. Chen et al. [146] limit the solution space when using differential evolutionary algorithms (also for ViTs and MLP).

Wang et al. [140] leverage geometric information from the data and optimize perturbations in the low-frequency space to reduce dimensionality, further enhancing query efficiency.

c) Transfer-based Attack and Adversarial Transferability. Underlying Reasons: Adversarial perturbations crafted to deceive one model can often mislead other unseen models. This adversarial transferability underpins transfer-based attacks, which learn adversarial perturbations on a surrogate model to attack unknown models. Researchers have investigated various factors influencing transferability, including the linear nature of deep models [2], adversarial subspace similarities [156], decision boundaries alignment [72], [147], [154], [156], loss surface smoothness [154], the convolutional nature [151], skip connections [152], model smoothness [153], reliance on non-robust features [155], data distribution [156], [157], and gradient similarity [153], [158]. Notably, transferability is not always symmetric. Perturbations that transfer from model A to model B may not work in the reverse direction [151], [154].

d) Adversarial Transferability Enhancement: Various studies [159], [166], [188] show that gradient-based iterative attack methods [2], [20], [74], [131] tend to fall into poor local optima and overfit the surrogate model. To address this, various strategies have been introduced.

Data augmentation improves adversarial transferability by enhancing data diversity. Methods include resizing [159], [162], translation [201], scale [166], adding random noise [205], spectrum transformation [206], data mix [38], [207], multi-path augmentation [160], adaptive transformation [208], hybrid transformation [209], block-wise transformation [161], [210], and style transformation [211].

Ensemble-based techniques use multiple models as the attack target [72], [188], [201], [212]. The rationale is that perturbations effective against several models are more likely to generalize to more models. Vanilla methods [72], [188], [201] average the outputs of all models uniformly, without considering the gradient variance across different models. This can lead to suboptimal solutions, where perturbations overfit to the ensemble as a whole. To avoid this, solutions include reducing the gradient variance [212], leveraging adaptive fusion [213], promoting loss landscape flatness and proximity to the local optima of each ensemble model [214]. There are also specialized ensemble techniques. For instance, Fang et al. [162] randomly modify backpropagation to achieve similar effects as the ensemble. Zhao et al. [163] exploit two models with significant discrepancies to construct an ensemble.

Momentum-based methods [164], [165], [166], [188], [215] accumulate gradients over time rather than relying solely on the current gradient. This helps stabilize update directions and

prevents the optimization process from falling into poor local optima. Vanilla methods simply accumulate the gradients of data points along the optimization path [188]. Recently, Wang et al. [164] additionally accumulate the gradients sampled in the direction of the previous iteration's gradient. Wang et al. [164], [215] fine-tune the gradient variance from the previous iteration. Long et al. [165] explore adaptive momentum strategies. Lin et al. [166] utilize Nesterov accelerated gradient as an enhanced momentum.

Architecture-oriented methods enhance adversarial transferability by focusing on the inner architectures of DNNs. Some approaches modify backpropagation to emphasize critical components. For instance, Wu et al. [152] discover that the skip connections are more vulnerable than other components and leverage gradients from skip connections to craft more transferable adversarial examples. Guo et al. [216] introduce linear backpropagation to skip nonlinear activations to encourage the linearity of DNNs. Xu et al. [167] modify the backpropagation path of convolution by structurally parameterizing and searching for optimal paths. Benz et al. [86] indicate that batch normalization makes the model more reliant on non-robust features and early stopping indeed helps significantly improve transferability. Furthermore, some works [168], [169], [217], [218], [219] learn perturbations by disturbing the intermediate feature instead of the output logit. Wu et al. [220], [221] discover that models commonly make decisions based on critical features and propose to craft adversarial noise to destroy critical features.

Some methods focus on finding *proper substitute models*, such as auto-encoding models [183], stylized networks [170], diffusion models [47], adversarially trained models [153], [171], models trained with norm-based regularization and gradient regularization [222]. Others employ *distribution-oriented approaches* to enhance transferability by shifting data away from its original distribution towards the target distribution [104], [157].

e) Cross-domain/modality Transferability: Recent studies learn image perturbations to affect video models by considering the low-level feature similarity between image and video models [172], [223], incorporating temporal information [224], and leveraging both global and local characteristics of video data [225]. Additionally, some explore cross-domain transferability by disrupting features in low-level [226], mid-level [226], intermediate and final layers [173]. Li et al. [174] exploit attention region commonality across different models and data transformation to encourage cross-domain cross-architecture transferability. Yang et al. [175] contrast clean and adversarial examples in the frequency domain by exploiting domain variant and invariant components. Wang et al. [171] employ a specific substitute model trained from scratch using contrastive learning.

f) Cross-task Transferability: Real-world systems often handle multiple tasks or utilize an ensemble of detection mechanisms, making it essential to examine transferability across various tasks. Cross-task transferability is limited by task-specific loss functions and models. To handle this, Lu et al. [227], [228] focus on the internal feature maps, which are independent of target-specific loss functions and models. Luo et al. [229]

propose to facilitate cross-prompt transferability across vision-language models by learning adversarial perturbations to make models to generate the target text "unknown". *Downstream-agnostic attack* is a specific type of cross-task attack. Given the prevalence of fine-tuning pre-trained encoders, it raises the question of whether their vulnerabilities persist in downstream models. These attacks are executed without knowledge of supervision signals or downstream tasks. Ban et al. [230] propose universal adversarial perturbations (UAPs) by perturbing low-level neuron activations, based on the observation that lower-layer parameters are less likely to change during fine-tuning. Zhou et al. [231] generate UAPs by altering the texture information of images, specifically high-frequency components. Some studies explore multi-modal downstream-agnostic attacks by disturbing multi-modal interactions [176], [178], increasing data diversity [176], [177] and encouraging local utility of perturbations [177], [178].

4) Attack ViTs and Cross-Architecture Transferability. Current Studies and Underlying Reasons for Vulnerability: Similar to CNNs, ViTs are sensitive to adversarial perturbations [67], [90], [92], [95], with reasons discussed in Section III-A-3. Recent efforts focus on enhancing adversarial transferability. For example, some construct an ensemble models as the attacking target [232]. Some concentrate on inner components. For instance, Wang et al. [179] activate uncertain attention and perturb sensitive embeddings. Wei et al. [45] skip the gradient of attention during backpropagation and randomly select a subset of adversarial patches to optimize. Zhang et al. [180] find that the gradient of skip connections significantly influences transferability and add virtual dense connections to back-propagate more gradients from deeper blocks. Some consider model-generic information. For example, Zhang et al. [181] reduce the variance of the back-propagated gradient to prevent attacks from focusing on model-specific features and getting stuck in poor local optima. Wei et al. [182] identify and drop model-specific critical regions during learning perturbation to prevent overfitting.

Comparison and Cross-architecture Transferability between CNNs and ViTs: CNNs and ViTs differ in architecture and data processing. CNNs use convolutional operations to exploit local features and patterns, while ViTs take a sequence of data patches as the input and leverage self-attention mechanisms to capture their global relationships. Mahmood et al. [232], [233], [234], [235] show that ViTs do not offer additional robustness. Moreover, adversarial examples show low transferability across ViTs, Big Transfer models, and CNNs. This may be because different networks focus on different frequency patterns: CNNs emphasize high-frequency details, whereas ViTs, due to their self-attention mechanism, focus on low-frequency patterns [93], [236]. Low-frequency reliance in ViTs can cause feature collapse in deeper layers. Gao et al. [168] propose to accelerate the feature collapse in later layers to improve transferability across CNNs and ViTs. Similarly, Chen et al. [237] suppress unique biases towards low-frequency information, and target common biases to improve transferability. Zhang et al. [180] leverage virtual dense connections to back-propagate more gradients from deeper blocks. Ma et al. [238] discover that

integrated gradients of images are highly similar across different models and use integrated gradients to learn transferrable perturbations. Tang et al. [239] propose reweighting ensemble models using reinforcement learning to increase the diversity of adversarial perturbations. Li et al. [174] exploit the commonality of attention regions across different models to improve transferability.

5) *Non-Box Attack*: Existing methods solving the cases where both data and model information are unavailable resort to finding a special substitute model, such as auto-encoder [183], or applying specific perturbation patterns, e.g., repetitive lines [79].

6) *Attack Against Defense*: Many defense methods have been proposed to counter adversarial perturbations for model reliability [20], [99], [186], [240], [241]. Specifically, Athalye et al. [128] argue that many gradient-based defense methods are effective against iterative optimization attacks because they cause obfuscated gradients, preventing effective loss optimization. Techniques including backward pass differentiable approximation, expectation over transformation, and reparameterization are employed to circumvent defense. Tramer et al. [184] show that defense strategies can be evaded by *adaptive attacks*, which are tailored to defenses. Large-norm [242] or unconstrained perturbations [34] can more likely fail defense models. Li et al. [185] propose to attack the diffusion-based purification method by pushing the reconstructed samples away from the diffused samples at intermediate diffusion steps.

7) *Limitations of the Optimization Objective*: Gradient-based attacks, such as L-BFGS [1] and PGD [20], learn perturbations by maximizing the cross-entropy (CE) loss. However, two critical issues remain. First, the CE loss is *unbounded*, meaning that a single instance can dominate the optimization process. For instance, a single misclassified instance can lead to arbitrarily large loss values. The second is the *vanishing gradient* (a kind of gradient masking) [74], [243], which arises when the model's probability predictions for original data are near 1. When attacking such a model to learn perturbations, the CE loss can increase sharply, causing substantial changes in gradient norms and making it difficult to find an appropriate learning rate. To handle these, alternative loss functions have been proposed, like bounded CE loss [186], logit loss [74], logits ratio loss [243], and Poincaré distance metric loss [187]. Other solutions include gradient normalization [244] and momentum-based gradient accumulation [188]. Gradient masking also appears in ViTs due to floating-point underflow errors in the softmax computation of each attention block [189], [245]. To mitigate this, Yu et al. [245] scale the logits adaptively before the softmax operation. Jain et al. [189] propose finding the optimal scaling factors of pre-softmax outputs using gradient-based optimization.

8) *Imperceptibility*: Perturbations, especially noise-based ones, inevitably degrade image quality. To address this, several approaches strive to make perturbations imperceptible. One method uses a Purifier network to pre-process adversarial examples, bringing them closer to original data in the perceptual feature space [190]. Semantic perturbations that are more natural than noise-based ones are leveraged, such as in spatial [59], [66], [67], [85], colour [33], [34], [60], texture [35] and style

spaces [192]. Additionally, some leverage diffusion models to enhance imperceptibility [47], [48], [191].

9) *Diverse Perturbations. Beyond l_p -norm Perturbations*: The widely used l_p -norm constraint only considers geometric distances in the pixel space, often missing perceptual differences between images. In some cases, small perturbations can lead to large changes in l_p -norm distance while causing minimal semantic changes. For example, simple rotations or translations may result in significant l_p -norm variations without altering the image's semantics. Such perturbations may not effectively deceive human observers or transfer well across models, and their optimization can be time-consuming. To address the limitation, several studies explore other distance constraints, including CIEDE2000 [61], [62], SSIM [62], [63], LPIPS [62], [64], [246], Wasserstein distance [65], geodesic distance [66], and parameter size based distance [67].

Beyond Single or Dual-type Perturbations. Early attacks typically only support single or dual-type perturbations such as l_2 -norm [244] and/or l_∞ -norm perturbations [20], [132]. Recently, progress has been made to support multiple l_p -norm perturbations simultaneously [247], [248], [249].

10) *Perturbation Constraints*: In the digital world, small perturbations can confuse deep models, but they often fail in the physical environment. This is primarily because minor perturbations may be too subtle to be detected in real-world settings. To succeed in physical attacks, perturbations need to be either *large* or *unrestricted*. Some methods focus on creating adversarial patches that are constrained in area but not in magnitude [58], [69], [70] to evade different systems. Other unrestricted perturbations are modifications on spatial properties [59], [66], [67], [85], colour [33], [34], [60], and texture [35] to keep perturbation natural. However, modifying these attributes still influences the image's appearance. To address this, Chen et al. [47], [48], [191] leverage diffusion models to enhance the imperceptibility. Wang et al. [192] transfer the style from the reference image to create natural-looking adversarial images.

11) *Attack Multimodal Large Language Models (MLLMs)*: Many recent works have begun to investigate the robustness of multimodal large language models (MLLMs). Different from conventional adversarial attacks on image classification, which mainly aim to induce prediction errors, attacks on MLLMs primarily seek to manipulate model behaviors to generate targeted, unsafe, or policy-violating responses, commonly referred to as *targeted attacks* and *jailbreak attacks*. Despite these differences in attack objectives and output spaces, many adversarial generation strategies adopted in MLLM attacks are still largely inspired by classical adversarial attack methods developed for image classification. This suggests that conventional perturbation generation and robustness analysis techniques may continue to provide valuable insights for the future development of adversarial robustness research beyond traditional classification settings.

For example, in targeted attack settings, Dong et al. [250] conduct transfer-based attacks by generating adversarial images that maximize the feature discrepancy from the original images. Furthermore, Zhao et al. [251] propose a query-based universal attack based on Monte Carlo gradient estimation and evaluate the

robustness of MLLMs by integrating transfer-based and query-based attack mechanisms.

In jailbreak attacks, Wang et al. [252] leverage universal adversarial images to facilitate malicious textual prompts. Several subsequent studies further explore converting prohibited textual prompts into images to bypass safety alignment. For example, Gong et al. [253] transform harmful textual prompts into typography-based images. Similarly, Li et al. [254] conceal harmful keywords within typographic images and further employ diffusion models to generate malicious images. Yang et al. [255] divide malicious prompts into multiple sub-prompts, convert them into typography-based subimages, and combine them with irrelevant distractor images while using benign textual prompts to induce harmful responses from MLLMs. In addition, Jeong et al. [96] apply transformations to harmful images and pair them with general prompts to bypass safety alignment mechanisms. Furthermore, Shayegani et al. [256] conceal harmful content within adversarial images while combining them with benign textual prompts to conduct jailbreak attacks.

B. OP-Based Poisoning Attack

1) *Definition: Targeted poisoning attack* aims to manipulate model behaviours for specific test instances. Feature Collision attack (FCA) [16] is a representative method, which seeks an imperceptible perturbation ξ on a base class instance x_b to minimize its distance to a target class instance x_t in the feature space:

$$\min_{x_{adv}} \|f_{\theta}(\xi(x_b)) - f_{\theta}(x_t)\|_2^2 + \beta \cdot \|\xi(x_b) - x_b\|_2^2. \quad (5)$$

When injected into the training set, the poisoned instances can cause the decision boundary to shift, including the target instance in the base class. As a result, the model may misclassify the target instance during testing. FCA is effective in fine-tuning scenarios but is less effective in models trained from scratch. The bi-level optimization method is proposed to enable the case of training from scratch [15], [257]:

$$\min_{\xi} \mathcal{L}(f_{\hat{\theta}}, \mathcal{D}_{val}), \text{ s.t. } \hat{\theta} = \arg \min_{\theta} \mathcal{L}(\theta, \hat{\mathcal{D}}_{train}), \quad (6)$$

where $\mathcal{D}_{val} = \{(x_{val}, y_{val})\}_{i=1}^n$, $y_{val} \neq y_{true}$, $\hat{\mathcal{D}}_{train} = \mathcal{D}_{clean1} \cup \mathcal{D}_{adv}$, $\mathcal{D}_{adv} = \{(\xi(x_{clean2}), y_{adv})\}_{i=1}^m$, $M(\xi(x_{clean2}), x_{clean2}) \leq \epsilon$, $(x_{clean2}, y_{clean2}) \in \mathcal{D}_{clean2}$. $\mathcal{D}_{val}$ is the validation dataset. $\mathcal{D}_{clean1}$ and $\mathcal{D}_{clean2}$ are clean datasets. y_{true} is the true label of x_{val} . Methods include meta-learning [257] and gradient matching [269] are proposed to solve this intractable problem.

Backdoor/Trojan attack injects perturbations into training data, leading target models to associate triggers with target labels. Consequently, test samples containing the trigger will be misclassified as the target label. A notable challenge is how to make target models associate triggers with target labels [258]. Solutions involve making clean samples harder to learn [258] and selecting harder-to-learn samples [259] to facilitate the learning of triggers. Zeng et al. [260] propose to learn error-minimizing triggers, which have strong relations with target

labels:

$$\min_{\xi} \mathbb{E}_{(x_{tar}, y_{tar}) \sim \mathcal{D}_{tar}} [\ell(f_{\theta}(\xi(x_{tar})), y_{tar})], \quad (7)$$

where $\mathcal{D}_{tar}$ is the dataset that contains target class samples.

Another solution solves a bi-level optimization problem [261], [262], [263], [264]:

$$\min_{\xi} f_{\hat{\theta}}(\xi(x_{val})) = y_t, \text{ s.t. } \hat{\theta} = \arg \min_{\theta} \mathcal{L}(\theta, \hat{\mathcal{D}}_{train}), \quad (8)$$

where $(x_{val}, y_{val}) \in \mathcal{D}_{val}$, $\mathcal{D}_{val}$ is the validation dataset. y_{adv} is the true label or another label. $y_t \neq y_{val}$. $\hat{\mathcal{D}}_{train} = \mathcal{D}_{clean1} \cup \mathcal{D}_{adv}$, $\mathcal{D}_{adv} = \{(\xi(x_{clean2}), y_{adv})\}_{i=1}^m$, $M(\xi(x_{clean2}), x_{clean2}) \leq \epsilon$, $(x_{clean2}, y_{clean2}) \in \mathcal{D}_{clean2}$. $\mathcal{D}_{clean1}$ and $\mathcal{D}_{clean2}$ are clean datasets. In addition to l_p -norm noise-based perturbations, other types, e.g., reflection-based perturbations [280], and colour perturbations [289] are used as backdoor triggers. In addition to l_p -norm noise-based perturbations, other types, e.g., reflection-based perturbations [280], and colour perturbations [289] are used as backdoor triggers.

Untargeted (a.k.a. Availability, Delusive, Indiscriminate) attack aims to degrade the model's performance on unseen clean data. Munoz et al. [15] proposes poisoning a portion of the dataset by bi-level optimization:

$$\max_{\xi} \mathcal{L}(f_{\hat{\theta}}, \mathcal{D}_{val}), \text{ s.t. } \hat{\theta} = \arg \min_{\theta} \mathcal{L}(\theta, \hat{\mathcal{D}}_{train}), \quad (9)$$

where $\mathcal{D}_{val} = \{(x_{val}, y_{val})\}_{i=1}^n$ is the validation dataset. y_{adv} is the true label or another label. $\hat{\mathcal{D}}_{train} = \mathcal{D}_{clean1} \cup \mathcal{D}_{adv}$, $\mathcal{D}_{adv} = \{(\xi(x_{clean2}), y_{adv})\}_{i=1}^m$, $M(\xi(x_{clean2}), x_{clean2}) \leq \epsilon$, $(x_{clean2}, y_{clean2}) \in \mathcal{D}_{clean2}$. $\mathcal{D}_{clean1}$ and $\mathcal{D}_{clean2}$ are clean datasets.

Recent works explore poisoning the entire dataset [265], [272], [279]. For instance, Fowl et al. [97] propose error-maximizing perturbations while other studies focus on error-minimizing perturbations [97], [265], [266] to accelerate model convergence, causing models to fail in learning meaningful data information. These perturbations are effective in supervised learning settings. However, He et al. [272], [279] reveal that unsupervised learning methods, like contrastive learning (CL), can still recover valuable representations poisoned datasets. To counter this, He et al. [279] perform poisoning by targeting three critical components of CL methods: the contrastive learning objective, data augmentation, and the momentum encoder. Ren et al. [272] quantify linear separability through the class-wise separability discriminant.

To enable model-free perturbation generation, Yu et al. [77] propose synthesizing linearly separable perturbations without solving optimization problems, based on the observation that perturbations for availability attacks tend to be nearly linearly separable. Sadasivan et al. [76] employ convolutional filters to learn controlled class-wise perturbations.

2) *Transferability*: Transferability is challenging due to unavailable target models and the re-training uncertainty. Variations in initialization and algorithms for re-training can yield significantly different models and limit attack efficacy [257], [267], [269], [271]. To address this, solutions include building a polytope to encompass target data [267], [268], employing special surrogate models, e.g., ensemble [257], [269], and

CLIP [270]. Additionally, He et al. [271] leverage the concept of the sharpness of the loss landscape to optimize the poisoning effect on the worst-case re-trained model. Ren et al. [272] promote perturbations linearly separable for availability attacks.

3) *Downstream-Agnostic Attack*: Poisoning attacks traditionally focus on supervised learning, but there is growing interest in addressing unsupervised settings, where labels are unavailable. This scenario is increasingly practical due to the difficulty of obtaining large labelled datasets for training models. Self-supervised learning, such as contrastive learning (CL), is often used to pre-train feature extractors for downstream tasks. For targeted poisoning, Liu et al. [273] concatenate a target input with a reference image as the poisoning input. An encoder trained on poisoned data would associate the target image and reference as similar, causing downstream classifiers to misclassify the target image as the reference's label. For backdoor attacks, Saha et al. [274] inject triggers into images of target classes. Jia et al. [275] propose training a backdoored encoder that produces similar feature vectors for both reference inputs and clean images, as well as for reference inputs and triggered samples. Other approaches incorporate augmentation during learning, such as learning augmentation-resistant spectral perturbations [276]. To avoid creating poisoned samples that are easily detected, Tao et al. [277] propose minimizing the distributional difference between clean and poisoned data. Zhang et al. [278] maximize the similarity between images including reference objects and those containing triggers. For untargeted attacks, He et al. [279] target three key components that ensure the performance of current CL methods: the CL objective, data augmentations, and the momentum encoder.

4) *Imperceptibility*: Imperceptibility is a critical challenge in backdoor attacks, arising from the *label* and the *input*. Many attacks manipulate labels, making them conspicuous and easy to detect [17], [18], [258], [287]. To mitigate this, clean-label attacks have emerged [16], [280], [281], [282]. On the input side, visible triggers in pixel space often expose attacks to the human eye and defense mechanisms. A common approach to overcoming detection is by limiting perturbation magnitude through distance constraints [16], [267], [268], [281], [283] or reducing trigger amplitude [258]. Some methods avoid injecting explicit backdoor triggers. For example, Saha et al. [282] learn a secret trigger to make poisoned data close to the target data in the input space while resembling data with the explicit trigger in the feature space. Other approaches include utilizing image steganography [263], sample-specific triggers [261], [262], dynamic triggers [284], image reflections [280], warp-based triggers [287], image scaling attacks [285], watermarking-based triggers [288], style-based triggers [290] and intensity-based methods [281]. Moreover, instead of focusing on the image/spatial domain, some works address imperceptibility from the perspective of the frequency domain. This is because backdoor attacks often exhibit high-frequency artefacts, making them easier to detect. Proposed solutions include avoiding high-frequency artefacts [264] and mixing clean and poisoned data in the frequency domain [291].

5) *Label-Agnostic Attack*: Current availability attacks, such as [97], [265], typically hold a label-consistency assumption that

the model trainer and attacker share the same label knowledge. However, in reality, datasets would go through distinct labelling processes. For instance, the trainer might label images with fine-grained classes like *cat* and *dog*, while the attacker may assign a more general label, such as *animal*. To address this discrepancy, Zhang et al. [270] propose to create perturbations at the cluster level rather than the class level.

6) *Poisoning Against Defense*: Some defense methods, like adversarial training [265], [292], randomized smoothing [28], [101], [129], and backdoor detection [293], have proven effective against poisoning attacks. To counter these, some integrate defense during perturbation learning [266], [286]. Specifically for backdoor triggers, techniques like adaptive triggers [262], tailored triggers [287], and latent space perturbations [288] have been developed.

7) *Connection Between Evasion and Poisoning Attacks*: Perturbations for evasion attacks have proven effective for availability attacks [97], while poisoning attacks can assist evasion attacks against adversarial training [294]. The main reason may be that different perturbations can commonly create shortcuts to models, thus can be used to influencing models at different stages.

8) *Poisoning Against ViTs*: Doan et al. [295], [296] show that ViTs are also vulnerable to backdoor attacks developed for CNNs. Due to the patch processing nature, ViTs are especially vulnerable to path-based poisoning attacks. For instance, Zheng et al. [46] implement patch-wise backdoor attacks based on patch salience.

C. Input Pre-Processing for Defense Via OP

1) *Defense Against OP-Based Evasion Attack*: There are works proposed to purify adversarial examples by means of generative models [297] and diffusion models [298], [299], which can be seen as a way to add optimization-guided perturbations on data. This approach faces high computational costs for multi-step denoising, requiring further exploration.

D. Adversarial Training (AT) for Defense

Most AT methods target evasion attacks of optimization-guided perturbations. Unless specified, the following content will discuss these methods. AT for others will be explicitly covered in Sections IV-D-13 and IV-E.

1) *Definition*: AT involves a two-player game between the adversary and the defender. The adversary generates worst-case perturbations to maximize the original training loss, while the defender aims to maintain prediction consistency [19], [20]. A representative objective is:

$$\min_{\theta} \mathbb{E}_{(x_{\text{train}}, y_{\text{train}}) \sim \mathcal{D}_{\text{train}}} [\max_{\xi} \ell(f_{\theta}(\xi(x_{\text{train}})), y_{\text{train}})]. \quad (10)$$

AT can be categorized into single-step AT [300], [301], [302], [303] and multi-step AT [20], [21], [22] based on the number of iterations used to generate perturbations. Multi-step AT generally provides better robustness than single-step AT, as it creates stronger adversarial examples. However, this comes at the cost

of efficiency due to the need for multiple forward and backward propagations to optimize the perturbations.

2) *Overfitting*: Overfitting is a common issue in AT, where models tend to overfit adversarial examples, leading to a significant degradation in test performance after a certain point [303], [393]. It can be categorized into *catastrophic overfitting* [303] and *robust overfitting* [393]. Catastrophic overfitting typically occurs in single-step AT, where training with a weaker adversary, such as FGSM, leads to a notable drop in robustness against stronger adversaries, like PGD. Robust overfitting happens when training and testing with the same adversary, often observed in multi-step AT.

a) *Catastrophic Overfitting: Underlying reasons* for catastrophic overfitting include several factors. First, from the sample perspective, these include a lack of diversity in adversarial examples or insufficient data [20], [303], low quality of adversarial examples [304], [305], presence of abnormal [306] or misclassified examples [307], [308], fast-growing gradients [319], and large gradient norms [314]. The second is model-related factors, including violations of local linearity in deep networks [300], [309], gradient masking effects [310], aliasing effects from poor downsampling [320], the self-fitting phenomenon [311], reliance on non-robust features [312], and over-memorization of specific patterns [313]. The third is the loss, including distortion of the decision boundary [301], [314], and compromise of local linearity in the loss function [315], [316]. Catastrophic overfitting is often accompanied by outliers in loss convergence [317].

Solutions: Based on the potential causes, many solutions have been proposed. From the *data* perspective, De et al. [318] introduce unrestricted noise-based data augmentations to enhance diversity. Some propose to learn high-quality adversarial perturbations from the initialization such as sample-dependent adversarial initialization [304], and historical perturbation-guided initialization [305]. In the inner maximization process, the loss associated with certain adversarial examples can decrease, leading to distorted decision boundaries. To mitigate this, Lin et al. [306] regularize the output variation of abnormal adversarial examples. Directly minimizing standard adversarial risk on misclassified examples is infeasible [307], [308]. Existing solutions include adding a regularization term to emphasize these samples [308], or incorporating inverse adversarial perturbations [307]. For the gradient-related problems, Li et al. [319] propose subspace adversarial training to restrict gradient growth within a carefully selected subspace, which also prevents robust overfitting. Huang et al. [314] use an adaptive step size for inner maximization.

From the *model* perspective, some propose to promote the local linearity of models by aligning the gradients between perturbed and original samples [300] and learning latent adversarial perturbations [309]. Vivek et al. [310] propose using dropout probability scheduling to mitigate the gradient masking effect. Grabinski et al. [320] introduce FrequencyLowCut pooling to ensure aliasing-free down-sampling. He et al. [311] prevent models from recognising the self-information in adversarial examples to avoid self-fitting. Lin et al. [313] leverage data

augmentation and high-confidence pattern removal to avoid over-memorization of certain patterns.

From the *loss* perspective, Kim et al. [301] propose to learn an adaptive perturbation magnitude to prevent distortion of the decision boundary. Rocamora et al. [315], [316] regularize the linear approximation error to enforce local linearity. Zhao et al. [317] restrict the loss difference between adjacent epochs to balance convergence and smoothing.

b) *Robust Overfitting: Underlying reasons* include limited sample complexity or insufficient data [321], [322], treating data equally [322], [323], [324], [325], models' bias towards global structure features [326] or non-robust features [321], [338], over-confidence in adversarial examples [324], [327], [328], [329], [332], and the limited capability of networks [343]. It is also related to the flatness of the weight loss landscape [331], [332], [333], [334], the input loss landscape [335], and robust local features [326]. Architectural design [330] and the choices of activation functions [339] play a role as well.

Solutions: Considering the *limited sample complexity*, utilizing external data has proven to be effective for mitigating overfitting. Strategies include incorporating data from external datasets [336], unlabelled data [386], or generative samples [367]. Another is data augmentation, which alleviates robust overfitting by increasing data diversity. Methods include heuristic augmentation, e.g., Cutout and MixUp [337], [394], data-driven and learning-based augmentations, e.g., AutoAugment [395], Diffusion-based augmentation [49], [333]. In addition to diversity, the hardness of augmentation is also important [396]. It is noted that augmentation should generate samples similar to the train and test distributions. For example, generative models like BigGAN may fail if generated data diverge significantly from the train and test distributions [333]. Moreover, Lau et al. [337] propose to learn robust adversarial perturbations to alleviate overfitting.

Different types of data have varying impacts on model robustness. *Treating all data equally* may lead to overfitting. One example is attackable data. Due to insufficient model capacity, data near the decision boundary (a.k.a. attackable data) are often overlooked. Solutions include geometry-aware instance reweighting [323], and adaptive perturbation magnitudes [324]. The second is adversarial data, which have unique characteristics and contribute differently to robustness. Yu et al. [325] increase the loss of small-loss data to mitigate overfitting. Song et al. [322] leverage distance-aware label smoothing to assign different soft labels for adversarial data based on their positions in adversarial regions. Cai et al. [397], [398] propose training models first with weak adversarial examples and then strong ones. In addition, misclassified samples complicate decision boundary adjustments during training, increasing the risk of overfitting. To handle this, some methods customize adversarial perturbation generation for correct and misclassified samples [21].

Based on the observation that well-generalizing models usually have relatively *flat loss surfaces* at the final learned weights, many studies propose improving robust generalization by flattening the weight loss landscape through model

weight averaging [331], [332], [333] and adversarial weight perturbation [334]. Li et al. [335] explore the relationship between robust overfitting and the input loss landscape, arguing that standard training, specifically minimizing clean loss, can lead to overfitting. To address this, a regularization technique is proposed to smooth the loss landscape by penalizing variations in weighted logits along the adversarial direction.

Considering *models' bias towards global structure features*, Song et al. [326] propose learning robust local features that are more generalizable and incorporating them into adversarial training. Memorizing non-robust features can create shortcuts that make models vulnerable to attacks. Wang et al. [338] attribute this vulnerability to an imbalance in the minimax game, caused by learning rate decay, and propose regularizing the defender's capacity and strengthening the attacks to restore balance between the two players.

To avoid *over-confidence* in adversarial data, confidence calibration is proposed, including logit smoothing [332], label smoothing [324], [327], temporal ensembling [328], and logit alignment between adversarial and clean examples [329].

Nakkiran et al. [343] argue that adversarial generalization requires *more complex or high-capacity networks*. Madry et al. [20], [336], [340] indicate that increasing model size can improve robustness but does not fully address robust overfitting. Singla et al. [339] find that the *choices of activation functions* significantly affects overfitting. Non-smooth activations, such as LeakyReLU, with low "approximate" curvature, can contribute to overfitting elimination.

Many studies *combine different strategies* to address overfitting, including model weight averaging with heuristic augmentation [394], learning-based augmentations with heuristic methods [333], adaptive perturbations with adaptive label smoothing [324], and label smoothing with augmentation [321]. However, not all combinations are effective. For instance, combining early stopping with heuristics augmentations does not offer additional reductions in overfitting compared to early stopping alone [49], [393].

3) *Adversarial Robustness Enhancement*: Mitigating overfitting is essential for enhancing adversarial robustness, which has been discussed above. In addition to it, there are many strategies proposed.

From the *data* perspective, Hou et al. [341] find that adaptively upweight hard-class pairs (i.e., semantically-similar classes) can enhance adversarial robustness. He et al. [342] propose emphasizing potentially missing shape information within frequency constraints to mitigate the texture bias of DNNs, thereby improving adversarial robustness.

For the *model* perspective, leveraging more complex and high-capacity models can enhance robustness [20], [336], [340], [343]. Several studies focus on robust feature activations. For instance, recognizing that adversarial perturbations can introduce noise into the feature map, Xie et al. [344] introduce a new denoising block into the network. Bai et al. [149] observe that adversarial perturbations can activate redundant and low-contributing channels, leading to reduced robustness, and propose to suppress less useful channels and only explore robust

feature activations. On the contrary, Kim et al. [345] reuse non-robust features to extract useful information. Ensemble-based AT is another widely adopted approach [240], [399]. In addition, Jin et al. [346] enhance robustness by optimizing second-order statistics of weights, which tightens the PAC-Bayesian adversarial generalization bound.

From the *feature* perspective, Wang et al. [347] exploit robust features that are resistant to adversarial perturbations. Mao et al. [348] regularize the feature space through contrastive learning. Kuang et al. [349] maintain both local and global topology structures in clean and adversarial feature spaces. Kuang et al. [350] leverage distillation methods to optimize the information bottleneck, reducing the sensitivity of intermediate features to perturbations.

From the *loss* perspective, some approaches enhance adversarial robustness by using alternative loss functions instead of the standard zero-one loss function, such as smooth adversarial loss [351] and contrastive learning [352].

4) *Robust Fairness*: Traditional AT achieves average adversarial robustness over classes, treating them equally. However, Different classes may exhibit varying levels of robustness. Treating them equally can result in significant robustness disparities between classes, significantly impacting real-world applications. For example, in autonomous driving, the weakness in a specific class, like "human", can create safety hazards [356]. To handle this, some propose class-level adaptive perturbations [353], [354], [355], and reweighting strategies [353], [356]. In addition, Zhou et al. [357] theoretically prove that perturbation directions and bounds are essential for robustness fairness and propose to use both error-maximizing and error-minimizing adversarial examples.

5) *Robustness-Accuracy Trade-Off*: Many AT methods [20], [186], [244] often overlook standard accuracy, i.e., the performance on normal data, which limits their real-world applicability. Since normal and adversarial examples exhibit distinct patterns, training with adversarial examples can degrade normal performance. Tsipras et al. [22], [400] highlight an inherent conflict between adversarial robustness and standard accuracy, suggesting a trade-off between them. Jin et al. [358], [372] indicate that clean accuracy is closely tied to decision boundary geometry and loss space flatness. Rade et al. [358] find that AT significantly alters decision boundary geometry, leading to accuracy degradation. The robustness-accuracy trade-off guides many AT methods [2], [19], [21], [99], [358]. Early approaches to balancing the trade-off by training models with both clean and adversarial data [2], [99], [358].

Some works propose to *align clean and adversarial data* in terms of class-conditional feature distributions [359], ranking [360], and distributions [361] to facilitate the trade-off. In addition, Yu et al. [362] propose to capture shared information between clean and adversarial data while discarding view-specific information to maintain the trade-off.

Kurakin et al. [20], [99] identify a potential *label-leaking* phenomenon that could influence the robustness-accuracy trade-off. This occurs when adversarial examples are constructed using true labels with single-step methods like FGSM, leading to

models performing better on adversarial data than on original data. One reason is that using true labels may learn perturbations that make the adversarial examples easier for the model to recognize. Madry et al. [20] further explain that label leaking occurs because FGSM produces a limited set of adversarial examples, which the network can easily overfit. To deal with it, Zhang et al. [363] propose using feature scattering to generate adversarial examples independent of true labels. Wang et al. [364] employ targeted attacks and random uniform noise as initialization.

From the *data* perspective, a line of research focuses on leveraging external data. For instance, Gowal et al. [367] incorporate additional data. Rade et al. [358] generate helper examples to preserve the geometry of the decision boundary. Lau et al. [337] utilize data augmentation to create diverse data. To mitigate the impact of *misclassified samples* on robustness and accuracy, Dong et al. [307] introduce inverse adversarial examples from adversarial examples and match the prediction between them. Zhou et al. [357] utilize both adversarial and anti-adversarial examples, given that different samples can have different perturbation directions and bounds, and are beneficial for the trade-off. Others optimize data usage. Zhang et al. [323] propose reweighting data based on the difficulty of attacking them. Kanai et al. [368] demonstrate that less-important samples can lead to overfitting and suggest increasing logit margins for more significant samples. Zhang et al. [369] leverage mild adversarial data.

From the *model* perspective, Dong et al. [370] introduce randomization into networks by replacing some convolutional filters with random projection filters, which helps maintain the distances among samples. Wei et al. [371] observe that standard and robust-trained models with the same architecture exhibit distinct filter weight distributions, making it difficult to bridge this gap with a static network. They use a dynamic network that adjusts weights at the filter level through a dynamic weight sub-network, combined with a MixBN structure to handle varying feature distributions. Jin et al. [372] indicate that the flat loss landscape and minima usually indicate good generalization ability. Many works propose to smooth model weights by leveraging stochastic weight averaging [332], Taylor expansion of Gaussian noise [372], and scaled noise [373]. Zhu et al. [374] propose identifying non-robust-critical modules of the network, which would be fine-tuned and then linearly interpolated with the adversarially trained weights to obtain better weights. Gowda et al. [375] empirically verify that different layers of a model contribute differently to accuracy-robustness balance and robust overfitting. They propose to selectively update important layers. Some studies [340], [376], [377] highlight that batch normalization (BN) in DNNs fails to capture distinct statistics of clean and adversarial samples, limiting overall performance. To solve this, Xie et al. [376] introduce an additional BN to model adversarial and clean distributions separately. In addition to auxiliary BN, Xie et al. [340] apply group normalization to avoid reliance on batch dimension statistics. However, BN still struggles with modelling mixture distributions. To deal with it, Wang et al. [377] propose a normalizer-free approach that eliminates all BN layers. Xie et al. [401] argue that the non-smooth nature of

the ReLU activation function contributes to weak adversarial robustness, and introduce smooth approximations of ReLU, which enhance adversarial robustness without compromising standard accuracy. In addition to modifying internal components, some approaches use distinct models to handle clean and adversarial data separately [402], [403].

Other methods focus on enhancing the *learning strategy*. For example, Shao et al. [365] indicate that AT often causes models to focus excessively on sparse spatial regions, neglecting valuable discriminative information. To address this, they propose attention-based AT. Mao et al. [366] combine PGD with sound interval bound propagation using a gradient connector to improve both certified and standard accuracy.

Some works challenge the inevitability of the robustness-accuracy trade-off. For example, Yang et al. [404] empirically show that using locally Lipschitz functions can achieve both robustness and accuracy. Raghunathan et al. [405] theoretically attribute the trade-off to overparameterization, explaining it through generalization from finite data.

6) *Defense Against Patch Attack*: Robust methods against imperceptible whole-area perturbations can hardly defeat patch attacks. To address this, Naseer et al. [378] reduce gradient activity in the attacked regions with gradient smoothing. Xu et al. [379] perform two-stage AT to train a patch detector that identifies adversarial patches at the pixel level and replaces them with mean pixel values.

7) *Multi-Attack Robustness*: Conventional AT typically defends against a single perturbation type. However, real-world scenarios often involve multiple perturbations, requiring multi-attack robustness. Tramer et al. [380] highlight that robustness to one perturbation type does not guarantee resistance to others, such as transitioning from l_∞ to l_1, l_2 and spatial perturbations. One solution is training with diverse adversarial perturbations, either using the strongest or using a range of perturbations [380]. However, this may lead to imbalanced robustness, as independent training with different perturbations may not minimize the loss across all types [381]. To address this, Maini et al. [381] propose learning a single perturbation under multiple perturbation models. Hsiung et al. [382] perform attack-order scheduling to learn with diverse perturbations. Other methods introduce additional strategies when learning with a single perturbation type. For example, Stutz et al. [327] enhance model robustness to various perturbations, including $l_p, p = \{0, 1, 2, \infty\}$ robustness, adversarial frames, distal adversarial perturbations and heuristic perturbations by training exclusively with l_∞ adversarial perturbations and encouraging the model to produce low-confidence predictions. Laidlaw et al. [64] use perceptual adversarial perturbations constrained by LPIPS distance to improve robustness to l_2, l_∞ -norm, spatial, recolouring, and JPEG attacks. Dai et al. [383] incorporate variation regularization into perceptual AT to minimize variability in the feature extractor across different threats, including LPIPS-based, l_2, l_∞ -norm, spatial, recolouring attacks. Kim et al. [352] combine contrastive learning with AT to generalize l_∞ robustness to l_1, l_2 adversaries.

8) *Robust Pre-Training and Fine-Tuning*: Deep models traditionally require extensive labelled training data for competitive performance, which can be hard to obtain. To mitigate this, many

approaches leverage large-scale unlabeled data with unsupervised learning techniques, to develop pre-trained models that are subsequently fine-tuned for downstream tasks. However, pre-trained models are often vulnerable to adversarial perturbations, necessitating robust enhancements. Common solutions incorporate AT into pre-trained objectives [352], [384], [385].

Despite these advances, adversarial pre-training alone does not always guarantee robustness in downstream tasks. Adversarial fine-tuning, which follows adversarial pre-training, is crucial [385], [406]. Vanilla adversarial fine-tuning follows the conventional AT regime [352], [384], [385], [386]. Recent advancements address the divergence between clean and adversarial data and explore the roles of different components. For instance, Liu et al. [387] use distinct batch normalization (BN) layers to preserve BN statistics from both the pre-trained model and the downstream task. To manage divergent optimization directions when optimizing natural and adversarial objectives, Xu et al. [388] introduce an additional low-rank branch to handle clean data and optimize the pre-trained feature extractor on adversarial data. Zhou et al. [389] highlight domain shifts caused by differences in training data between pre-training and fine-tuning, as well as catastrophic knowledge forgetting. They propose a two-stage adversarial fine-tuning method, which first optimizes the pre-trained encoder and classifier separately and then selectively fine-tunes layers according to robustness.

9) *Adaptive Learning*: Perturbation magnitude is critical in AT. Small perturbations may not ensure robustness, while large ones can degrade accuracy. Additionally, data diversity and intrinsic robustness levels matter. Some examples are naturally closer to the decision boundary, and enforcing large margins on them may result in a distorted decision surface. This can lead to overfitting [301], [324], affect robustness [241] and influence the robustness-accuracy trade-off [241]. Therefore, applying a uniform perturbation magnitude across data may be ineffective. To address this, instance-level adaptive perturbations [241], [301], [324] and class-level adaptive perturbations are designed [353], [354], [355].

10) *Efficiency*: AT is its high computational cost, which requires multiple gradient descent computations to create adversarial images and compute gradients for network updates [20], [302], [390]. To address this, some methods minimize gradient calculations [302], [315], [390] and optimise sample selection [391]. Others leverage single-step AT with improved initialization [303], [304], [305] and lightweight surrogate models [392] to enhance efficiency.

11) *Comparison and Connection Between AT and RS*: RS offers certified robustness within a fixed perturbation bound but is limited by the size of manageable perturbations. In contrast, AT enhances empirical defense against stronger attacks without guaranteeing theoretical robustness. Combining these methods allows models to leverage the certified robustness of RS and the empirical strength of AT, enabling provable defense against larger and more diverse attacks. For instance, Sinha et al. [100] explore provable robustness from Wasserstein distributionally robust optimization by leveraging Wasserstein adversarial perturbations. Salman et al. [101] incorporate adversarial examples

into the training process to boost the provable l_2 robustness of smoothed classifiers. Yang et al. [102] augment adversarial examples with Gaussian noise to strengthen robustness certificates. Wang et al. [103] integrate AT-based smoothing with distributionally robust optimization to ensure distributional robustness.

12) *AT for ViTs and Comparison With CNNs*: Conventional AT methods seem less effective for ViTs compared to CNNs. Bai et al. [235], [236], [407] demonstrate that applying AT designed for CNNs to ViTs does not yield better robustness than large CNNs. Bai et al. [235] adversarially train DeiT-S using single-step PGD with strong data augmentation but still fail against PGD-5. They find that applying CNN-based AT recipes, such as using M-SGD with a step decay learning rate scheduler instead of adaptive optimizers like AdamW, diminishes both accuracy and robustness. Moreover, applying AT to ViTs faces the high computational cost associated with the self-attention mechanism [407]. To enable efficient and effective AT on ViTs, Wu et al. [407] drop the patches based on the attention information.

Unlike CNNs, which benefit from built-in inductive biases due to convolutional filters, ViTs lack these inherent properties and require large amounts of data and augmentations to learn them from scratch. However, the standard training recipe for ViTs, which emphasizes strong data augmentation, is often suboptimal for AT [235], [408]. Considering this, Bai et al. [235] progressively enhance augmentation strength to better warm up the training process. DeBenedetti et al. [408] use lighter data augmentation (e.g., random resizing, and cropping) along with linear warm-up for perturbation magnitude and high weight decay. Random weight initialization in ViTs is often inadequate for AT [409], making pre-trained initialization necessary. Training from scratch typically fails to converge optimally. While natural pre-training on large datasets aids AT, simply increasing dataset size doesn't guarantee improved robustness [409]. Gradient clipping, optional in standard training, is crucial for AT. Prolonged training can undermine final robustness [409]. SGD with a piecewise learning rate scheduler is a preferred approach for AT of ViTs [409].

13) *AT Against Poisoning Attack*: Gao et al. [410] show that AT's effectiveness against backdoor attacks depends on the type and budget of perturbations. AT with l_∞ perturbations [20] is effective against whole-image backdoor attacks. AT with spatial perturbations mitigates patch-based backdoor attacks but may slightly reduce accuracy. The effectiveness of AT against availability attacks has been verified in [265], [292].

E. Relation Between Different Defense

Relations between Different Adversarial Robustness to Evasion Attack: Engstrom et al. [67] show that adversarial geometric training cannot ensure robustness to l_p adversaries and robustness to l_∞ -norm bounded perturbations does not significantly affect spatial robustness.

Relations between Adversarial Robustness to Evasion Attack and Backdoor Attack: Weng et al. [416] discover that AT, which boosts model resilience to evasion attacks, also increase robustness to backdoor attacks. Niu et al. [418] further reveal two

TABLE V
REPRESENTATIVE BENCHMARKS FOR IMAGE PERTURBATION-BASED ATTACK AND DEFENSE EVALUATION

Benchmark	Threat Model	Dataset	Metrics	Limitations
ImageNet-C [31]	Common perturbation	ImageNet, CIFAR	Accuracy, mCE, Relative mCE	Synthetic and non-worst-case perturbations
ImageNet-P [31]	Sequential perturbation	ImageNet	mFR, mTSD	Synthetic and non-worst-case perturbations
Obfuscation RobustBench [126]	Evasion attack via image obfuscations	ImageNet	Robust accuracy, Attack success rate	Synthetic, and Primarily targets image obfuscation settings
RobustBench [411]	l_p -norm evasion attacks, $p \in \{2, \infty\}$	CIFAR-10, CIFAR-100, ImageNet	Clean accuracy, Robust accuracy	Limited to l_∞ and l_2 -norm robustness
AttackBench [412]	l_p -norm evasion attacks, $p \in \{0, 1, 2, \infty\}$	CIFAR-10, ImageNet	Optimality scores, Attack success rate, Robust accuracy	Limited to norm-bounded and gradient-based attacks
RealSafe [413]	l_p -norm evasion attacks, $p \in \{2, \infty\}$	CIFAR-10, ImageNet	Robust accuracy, ASR	Limited to l_2 -norm robustness
Poisoning-benchmark [414]	l_∞ -norm backdoor and poisoning attacks	CIFAR-10, TinyImageNet	Attack success rate, Clean accuracy, Defense success rate	Limited to l_p -norm robustness, Evaluation Metrics are limited to success rate, without stealthiness, transferability
BackdoorBench [416]	Backdoor attack	CIFAR-10, CIFAR-100, GTSRB, Tiny ImageNet	Clean accuracy, Robust accuracy, Attack success rate, Analysis tool: T-SNE, Shapley value, Grad-CAM, Frequency saliency map, Neuron activation	Limited numbers of attack and defense methods
Blackboxbench [415]	Query-based, Transfer-based evasion attacks	CIFAR-10, ImageNet	Adversarial Divergence, The ratio of AQN to ASR, MSE	Limited to norm-bounded adversarial perturbations, limited analysis of perceptual quality, stealthiness

insights: (1) implanting a backdoor in a model significantly impacts its adversarial examples, and (2) for a compromised model, adversarial examples for evasion attacks share features with triggered images. To address this, they propose a progressive purification method to erase backdoors and enhance robustness to evasion attacks.

Relations between Adversarial Robustness to OP and HP: Gilmer et al. [419] indicate that adversarial training also improve robustness to heuristic perturbations, and training on Gaussian noise moderately improves adversarial robustness. Kireev et al. [421] find that with a proper perturbation radius, l_p -AT can improve robustness to heuristic adversarial perturbations. On the contrary, Stimberg et al. [126] show that training on obfuscations does not give any robustness to l_p -norm attacks and l_p -norm adversarial training reduces the obfuscation robustness. In addition, certifying robustness against l_p -norm perturbations is not sufficient for attacks based on semantic transformation. The reason may be that semantic transformation-based perturbations incur large l_p -norm differences [118], [420]. Xu et al. [421] leverage adversarial perturbations to encourage style-invariant learning.

F. Efficiency and Practical Considerations

Practical deployment requires a unified analysis of efficiency, scalability, and data dependency. For heuristic perturbation (HP)-based attacks and defenses, no explicit optimization is required, leading to high efficiency but often at the cost of effectiveness. In contrast, optimization-based (OP) attacks involve higher computational overhead.

For evasion attacks, transfer-based methods incur costs during surrogate model training, while query-based methods depend on the number of queries, which is influenced by factors such as data dimensionality. As a result, query-based attacks are generally less efficient than transfer-based ones. For poisoning attacks, efficiency is typically lower than evasion attacks due to the need for bi-level optimization and training-time data manipulation. Moreover, untargeted poisoning attacks often incur higher cost than targeted or backdoor attacks, as they require perturbing a larger portion of the dataset.

On the defense side, adversarial training, randomized smoothing, and input processing introduce substantial computational and data overhead. Adversarial training requires extensive data augmentation with diverse perturbations, randomized smoothing incurs repeated inference with perturbed inputs, and input

preprocessing methods rely on a large number of data perturbations during training or inference.

In terms of scalability, many optimization-based attacks and defenses face challenges when applied to large-scale or high-dimensional systems due to increased computational and query costs. From a data perspective, defense methods typically require large-scale, high-quality data, while attack methods often rely on datasets or extensive data manipulation, reflecting different levels of data dependency.

G. Benchmarking of Perturbation Robustness

A summary of representative robustness benchmarks is presented in Table V. Despite the progress achieved, current perturbation robustness evaluation frameworks still exhibit several important limitations. First, most existing benchmarks primarily focus on synthetic perturbations and lack evaluations on natural adversarial examples and realistic distribution shifts, which may limit the practical assessment of model robustness. Second, many benchmarks only consider limited perturbation types, while insufficiently covering diverse perturbation settings, including l_p -norm-bounded perturbations, geometric/spatial perturbations, and perceptual perturbations. Moreover, robustness evaluations are mainly conducted during a single stage, such as inference, while robustness analysis across different stages of the learning pipeline remains underexplored. Furthermore, existing evaluation metrics mainly focus on final prediction outcomes and fail to adequately investigate perturbation characteristics and their broader influence on model behavior, such as perceptual quality, perturbation stealthiness, changes in data relationships, and decision boundary variations. Consequently, existing benchmarks remain insufficient for comprehensively characterizing model vulnerabilities and robustness limitations under complex real-world conditions.

V. CONCLUSION

In this paper, we present a comprehensive survey of data perturbations - focusing on the image domain - and their applications in enhancing the robustness of deep learning models. We examine the taxonomy of these perturbations, their generation methods, and the key challenges and solutions associated with them. Furthermore, we explore the impact of perturbations on model robustness and provide insights into the underlying mechanisms and causes. This survey offers readers a consolidated

overview of current research on image perturbations, contributing to a deeper understanding of the field and supporting the development of robust and trustworthy systems. A discussion of future research directions is provided in the supplementary material.

REFERENCES

- [1] C. Szegedy et al., "Intriguing properties of neural networks," in *Proc. Int. Conf. Learn. Representations*, 2013.
- [2] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," in *Proc. Int. Conf. Learn. Representations*, 2014.
- [3] X. Wang et al., "Revisiting adversarial attacks on graph neural networks for graph classification," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 5, pp. 2166–2178, May 2024.
- [4] Y. Zhu, L. Tong, G. Li, X. Luo, and K. Zhou, "FocusedCleaner: Sanitizing poisoned graphs for robust GNN-based node classification," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 6, pp. 2476–2489, Jun. 2024.
- [5] P.-F. Zhang, Y. Li, Z. Huang, and H. Yin, "Privacy protection in deep multi-modal retrieval," in *Proc. Int. ACM SIGIR Conf. Res. Develop. Inform. Retrieval*, 2021, pp. 634–643.
- [6] J. Li, R. Ji, H. Liu, X. Hong, Y. Gao, and Q. Tian, "Universal perturbation attack against image retrieval," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2019, pp. 4898–4907.
- [7] T. Wang, F. Li, L. Zhu, J. Li, Z. Zhang, and H. T. Shen, "Invisible black-box backdoor attack against deep cross-modal hashing retrieval," *ACM Trans. Inf. Syst.*, vol. 42, 2024, Art. no. 111.
- [8] L. Jiang, X. Ma, S. Chen, J. Bailey, and Y.-G. Jiang, "Black-box adversarial attacks on video recognition models," in *Proc. 27th ACM Int. Conf. Multimedia*, 2019, pp. 864–872.
- [9] H. Zhang, L. Zhu, Y. Zhu, and Y. Yang, "Motion-excited sampler: Video adversarial attack with sparked prior," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 240–256.
- [10] Z. Ye et al., "Speed master: Quick or slow play to attack speaker recognition," in *Proc. Conf. Assoc. Advance. Artif. Intell.*, 2025, pp. 22137–22145.
- [11] W. Yao, J. Yang, Y. He, J. Liu, and W. Wen, "Imperceptible rhythm backdoor attacks: Exploring rhythm transformation for embedding undetectable vulnerabilities on speech recognition," *Neurocomputing*, vol. 614, 2025, Art. no. 128779.
- [12] J. Liu et al., "Aliasing black box adversarial attack with joint self-attention distribution and confidence probability," *Expert Syst. Appl.*, vol. 214, 2023, Art. no. 119110.
- [13] M. Alzantot, Y. Sharma, A. Elgohary, B.-J. Ho, M. Srivastava, and K.-W. Chang, "Generating natural language adversarial examples," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2018, pp. 2890–2896.
- [14] N. Papernot, P. McDaniel, I. Goodfellow, S. Jha, Z. B. Celik, and A. Swami, "Practical black-box attacks against machine learning," in *Proc. ACM Asia Conf. Comput. Secur.*, 2017, pp. 506–519.
- [15] L. Muñoz-González et al., "Towards poisoning of deep learning algorithms with back-gradient optimization," in *Proc. 10th ACM Workshop Artif. Intell. Secur.*, 2017, pp. 27–38.
- [16] A. Shafahi et al., "Poison frogs! Targeted clean-label poisoning attacks on neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2018, pp. 6106–6116.
- [17] Y. Liu et al., "Trojaning attack on neural networks," in *Proc. Annu. Netw. Distrib. Syst. Secur. Symp.*, 2018.
- [18] X. Chen, C. Liu, B. Li, K. Lu, and D. Song, "Targeted backdoor attacks on deep learning systems using data poisoning," 2017, *arXiv:1712.05526*.
- [19] H. Kannan, A. Kurakin, and I. Goodfellow, "Adversarial logit pairing," 2018, *arXiv:1803.06373*.
- [20] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," in *Proc. Int. Conf. Learn. Representations*, 2017.
- [21] Y. Wang, D. Zou, J. Yi, J. Bailey, X. Ma, and Q. Gu, "Improving adversarial robustness requires revisiting misclassified examples," in *Proc. Int. Conf. Learn. Representations*, 2019.
- [22] H. Zhang, Y. Yu, J. Jiao, E. Xing, L. El Ghaoui, and M. Jordan, "Theoretically principled trade-off between robustness and accuracy," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 7472–7482.
- [23] R. Geirhos, P. Rubisch, C. Michaelis, M. Bethge, F. A. Wichmann, and W. Brendel, "ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness," in *Proc. Int. Conf. Mach. Learn.*, 2018.
- [24] Z. Zhong, L. Zheng, G. Kang, S. Li, and Y. Yang, "Random erasing data augmentation," in *Proc. Conf. Assoc. Advance. Artif. Intell.*, 2020, pp. 13001–13008.
- [25] A. Azulay and Y. Weiss, "Why do deep convolutional networks generalize so poorly to small image transformations?," *J. Mach. Learn. Res.*, vol. 20, pp. 1–25, 2019.
- [26] D. Hendrycks et al., "The many faces of robustness: A critical analysis of out-of-distribution generalization," in *Proc. Int. Conf. Comput. Vis.*, 2021, pp. 8340–8349.
- [27] M. Lecuyer, R. Atliedakis, D. Hsu, and S. Jana, "Certified robustness to adversarial examples with differential privacy," in *Proc. IEEE Symp. Secur. Privacy*, 2019, pp. 656–672.
- [28] J. Cohen, E. Rosenfeld, and Z. Kolter, "Certified adversarial robustness via randomized smoothing," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 1310–1320.
- [29] G. Yang, T. Duan, J. E. Hu, H. Salman, I. Razenshteyn, and J. Li, "Randomized smoothing of all shapes and sizes," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 10693–10705.
- [30] S. Ugare, T. Suresh, D. Banerjee, G. Singh, and S. Misailovic, "Incremental randomized smoothing certification," in *Proc. Int. Conf. Learn. Representations*, 2023.
- [31] D. Hendrycks and T. Dietterich, "Benchmarking neural network robustness to common corruptions and perturbations," in *Proc. Int. Conf. Learn. Representations*, 2018.
- [32] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2012, pp. 1097–1105.
- [33] H. Hosseini and R. Poovendran, "Semantic adversarial examples," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshop*, 2018, pp. 1614–1619.
- [34] A. S. Shamsabadi, R. Sanchez-Matilla, and A. Cavallaro, "ColorFool: Semantic adversarial colorization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 1151–1160.
- [35] A. Bhattad, M. J. Chong, K. Liang, and B. Li, "Unrestricted adversarial examples via semantic manipulation," in *Proc. Int. Conf. Learn. Representations*, 2019.
- [36] C. Guo, M. Rana, M. Cisse, and L. van der Maaten, "Countering adversarial images using input transformations," in *Proc. Int. Conf. Learn. Representations*, 2018.
- [37] D. Hendrycks, N. Mu, E. D. Cubuk, B. Zoph, J. Gilmer, and B. Lakshminarayanan, "AugMix: A simple data processing method to improve robustness and uncertainty," in *Proc. Int. Conf. Learn. Representations*, 2019.
- [38] X. Wang, X. He, J. Wang, and K. He, "Admix: Enhancing the transferability of adversarial attacks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 16138–16147.
- [39] A. Chakraborty, M. Alam, V. Dey, A. Chattopadhyay, and D. Mukhopadhyay, "Adversarial attacks and defences: A survey," 2018, *arXiv:1810.00069*.
- [40] S. Han, C. Lin, C. Shen, Q. Wang, and X. Guan, "Interpreting adversarial examples in deep learning: A review," *ACM Comput. Surv.*, vol. 55, pp. 1–38, 2023.
- [41] Z. Tian, L. Cui, J. Liang, and S. Yu, "A comprehensive survey on poisoning attacks and countermeasures in machine learning," *ACM Comput. Surv.*, vol. 55, no. 8, pp. 1–35, 2022.
- [42] Y. Li, Y. Jiang, Z. Li, and S.-T. Xia, "Backdoor learning: A survey," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 1, pp. 5–22, Jan. 2024.
- [43] H. Sun, T. Zhu, Z. Zhang, D. Jin, P. Xiong, and W. Zhou, "Adversarial attacks against deep generative models on data: a survey," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 4, pp. 3367–3388, Apr. 2023.
- [44] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. Int. Conf. Learn. Representations*, 2015.
- [45] Z. Wei, J. Chen, M. Goldblum, Z. Wu, T. Goldstein, and Y.-G. Jiang, "Towards transferable adversarial attacks on vision transformers," in *Proc. Conf. Assoc. Advance. Artif. Intell.*, 2022, pp. 2668–2676.
- [46] M. Zheng, Q. Lou, and L. Jiang, "TrojViT: Trojan insertion in vision transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 4025–4034.
- [47] J. Chen, H. Chen, Y. Zhang, Z. Zou, and Z. Shi, "Diffusion models for imperceptible and transferable adversarial attack," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 2, pp. 961–977, Feb. 2025.
- [48] X. Chen, X. Gao, J. Zhao, K. Ye, and C.-Z. Xu, "AdvDiffuser: Natural adversarial example synthesis with diffusion models," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 4539–4549.

- [49] Z. Wang, T. Pang, C. Du, M. Lin, W. Liu, and S. Yan, "Better diffusion models further improve adversarial training," in *Proc. Int. Conf. Mach. Learn.*, 2023, pp. 36246–36263.
- [50] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 8748–8763.
- [51] J. Li, D. Li, C. Xiong, and S. Hoi, "BLIP: Bootstrapping language-image pre-training for unified vision-language understanding and generation," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 12888–12900.
- [52] R. Bommasani et al., "On the opportunities and risks of foundation models," 2021, *arXiv:2108.07258*.
- [53] J. Wei et al., "Emergent abilities of large language models," *Trans. Mach. Learn. Res.*, 2022.
- [54] N. Akhtar and A. Mian, "Threat of adversarial attacks on deep learning in computer vision: A survey," *IEEE Access*, vol. 6, pp. 14410–14430, 2018.
- [55] S. Zhou, C. Liu, D. Ye, T. Zhu, W. Zhou, and P. S. Yu, "Adversarial attacks and defenses in deep learning: From a perspective of cybersecurity," *ACM Comput. Surv.*, vol. 55, no. 8, pp. 1–39, 2022.
- [56] N. Karunanayake, R. Gunawardena, S. Seneviratne, and S. Chawla, "Out-of-distribution data: An acquaintance of adversarial examples—a survey," *ACM Comput. Surv.*, vol. 57, no. 8, pp. 1–40, 2025.
- [57] A. Chakraborty, M. Alam, V. Dey, A. Chattopadhyay, and D. Mukhopadhyay, "A survey on adversarial attacks and defences," *CAAI Trans. Intell. Technol.*, vol. 6, no. 1, pp. 25–45, 2021.
- [58] M. Sharif, S. Bhagavatula, L. Bauer, and M. K. Reiter, "Accessorize to a crime: Real and stealthy attacks on state-of-the-art face recognition," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, 2016, pp. 1528–1540.
- [59] F. Wang, P. Xu, W. Ruan, and X. Huang, "Towards verifying the geometric robustness of large-scale neural networks," in *Proc. AAAI Conf. Artif. Intell.*, 2023, vol. 37, pp. 15197–15205.
- [60] S. Yuan, Q. Zhang, L. Gao, Y. Cheng, and J. Song, "Natural color fool: Towards boosting black-box unrestricted attacks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 7546–7560.
- [61] Z. Zhao, Z. Liu, and M. Larson, "Towards large yet imperceptible adversarial image perturbations with perceptual color distance," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 1039–1048.
- [62] J. Rony, E. Granger, M. Pedersoli, and I. Ben Ayed, "Augmented lagrangian adversarial attacks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 7738–7747.
- [63] D. Gragnaniello, F. Marra, L. Verdoliva, and G. Poggi, "Perceptual quality-preserving black-box attack against deep learning image classifiers," *Pattern Recognit. Lett.*, vol. 147, pp. 142–149, 2021.
- [64] C. Laidlaw, S. Singla, and S. Feizi, "Perceptual adversarial robustness: Defense against unseen threat models," in *Proc. Int. Conf. Learn. Representations*, 2021.
- [65] E. Wong, F. Schmidt, and Z. Kolter, "Wasserstein adversarial examples via projected sinkhorn iterations," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 6808–6817.
- [66] C. Kanbak, S.-M. M-D, and P. Frossard, "Geometric robustness of deep networks: Analysis and improvement," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 4441–4449.
- [67] L. Engstrom, B. Tran, D. Tsipras, L. Schmidt, and A. Madry, "Exploring the landscape of spatial robustness," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 1802–1811.
- [68] J. Su, D. V. Vargas, and K. Sakurai, "One pixel attack for fooling deep neural networks," *IEEE Trans. Evol. Comput.*, vol. 23, no. 5, pp. 828–841, Oct. 2019.
- [69] T. B. Brown, D. Mané, A. Roy, M. Abadi, and J. Gilmer, "Adversarial patch," 2017, *arXiv:1712.09665*.
- [70] K. Xu et al., "Adversarial t-shirt! Evading person detectors in a physical world," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 665–681.
- [71] N. Papernot, P. McDaniel, S. Jha, M. Fredrikson, Z. B. Celik, and A. Swami, "The limitations of deep learning in adversarial settings," in *Proc. IEEE Eur. Symp. Secur. Privacy*, 2016, pp. 372–387.
- [72] Y. Liu, X. Chen, C. Liu, and D. Song, "Delving into transferable adversarial examples and black-box attacks," in *Proc. Int. Conf. Learn. Representations*, 2016.
- [73] T. Zheng, C. Chen, and K. Ren, "Distributionally adversarial attack," in *Proc. AAAI Conf. Artif. Intell.*, 2019, vol. 33, pp. 2253–2260.
- [74] N. Carlini and D. Wagner, "Towards evaluating the robustness of neural networks," in *Proc. IEEE Symp. Secur. Privacy*, 2017, pp. 39–57.
- [75] S.-M. Moosavi-Dezfooli, A. Fawzi, and P. Frossard, "DeepFool: A simple and accurate method to fool deep neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 2574–2582.
- [76] V. S. Sadasivan, M. Soltanolkotabi, and S. Feizi, "CUDA: Convolution-based unlearnable datasets," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 3862–3871.
- [77] D. Yu, H. Zhang, W. Chen, J. Yin, and T.-Y. Liu, "Availability attacks create shortcuts," in *Proc. ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, 2022, pp. 2367–2376.
- [78] S.-M. Moosavi-Dezfooli, A. Fawzi, O. Fawzi, and P. Frossard, "Universal adversarial perturbations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 1765–1773.
- [79] C. Zhang, P. Benz, A. Karjauv, and I. S. Kweon, "Data-free universal adversarial perturbation and black-box attack," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 7848–7857.
- [80] S. Dodge and L. Karam, "A study and comparison of human and deep learning recognition performance under visual distortions," in *Proc. 26th Int. Conf. Comput. Commun. Netw.*, pp. 1–7.
- [81] I. Vasiljevic, A. Chakrabarti, and G. Shakhnarovich, "Examining the impact of blur on recognition by convolutional networks," 2016, *arXiv:1611.05760*.
- [82] R. Zhang, "Making convolutional networks shift-invariant again," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 7324–7334.
- [83] X. Zou, F. Xiao, Z. Yu, Y. Li, and Y. J. Lee, "Delving deeper into anti-aliasing in convnets," *Int. J. Comput. Vis.*, vol. 131, no. 1, pp. 67–81, 2023.
- [84] M. T. Hossain, S. W. Teng, G. Lu, M. A. Rahman, and F. Soheli, "Anti-aliasing deep image classifiers using novel depth adaptive blurring and activation function," *Neurocomputing*, vol. 536, pp. 164–174, 2023.
- [85] H. Michaeli, T. Michaeli, and D. Soudry, "Alias-free convnets: Fractional shift invariance via polynomial activations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 16333–16342.
- [86] P. Benz, C. Zhang, and I. S. Kweon, "Batch normalization increases adversarial vulnerability and decreases adversarial transferability: A non-robust feature perspective," in *Proc. Int. Conf. Comput. Vis.*, 2021, pp. 7818–7827.
- [87] Y. Qin, C. Zhang, T. Chen, B. Lakshminarayanan, A. Beutel, and X. Wang, "Understanding and improving robustness of vision transformers through patch-based negative augmentation," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 16276–16289.
- [88] S. Qian, H. Shao, Y. Zhu, M. Li, and J. Jia, "Blending anti-aliasing into vision transformer," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 5416–5429.
- [89] R. A. Rojas-Gomez, T.-Y. Lim, M. N. Do, and R. A. Yeh, "Making vision transformers truly shift-equivariant," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 5568–5577.
- [90] S. Bhojanapalli, A. Chakrabarti, D. Glasner, D. Li, T. Unterthiner, and A. Veit, "Understanding robustness of transformers for image classification," in *Proc. Int. Conf. Comput. Vis.*, 2021, pp. 10231–10241.
- [91] J. Gu, V. Tresp, and Y. Qin, "Are vision transformers robust to patch perturbations?," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 404–421.
- [92] Y. Guo, D. Stutz, and B. Schiele, "Improving robustness of vision transformers by reducing sensitivity to patch corruptions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 4108–4118.
- [93] M. M. Naseer, K. Ranasinghe, S. H. Khan, M. Hayat, F. Shahbaz Khan, and M.-H. Yang, "Intriguing properties of vision transformers," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 23296–23308.
- [94] M. Moayeri, P. Pope, Y. Balaji, and S. Feizi, "A comprehensive study of image classification model sensitivity to foregrounds, backgrounds, and visual attributes," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 19087–19097.
- [95] X. Mao et al., "Towards robust vision transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 12032–12041.
- [96] J. Jeong, S. Bae, Y. Jung, J. Hwang, and E. Yang, "Playing the fool: Jailbreaking LLMs and multimodal LLMs with out-of-distribution strategy," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2025, pp. 29937–29946.
- [97] L. Fowl, M. Goldblum, W. Czaja, and T. Goldstein, "Adversarial examples make strong poisons," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 30339–30351.
- [98] T. Xu, Y. Li, Y. Jiang, and S.-T. Xia, "Batt: Backdoor attack with transformation-based triggers," in *Proc. Int. Conf. Acoust., Speech Signal Process.*, 2023, pp. 1–5.
- [99] A. Kurakin, I. Goodfellow, and S. Bengio, "Adversarial machine learning at scale," in *Proc. Int. Conf. Learn. Representations*, 2016.

- [100] A. Sinha, H. Namkoong, R. Volpi, and J. Duchi, "Certifying some distributional robustness with principled adversarial training," in *Proc. Int. Conf. Learn. Representations*, 2017.
- [101] H. Salman et al., "Provably robust deep learning via adversarially trained smoothed classifiers," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 11292–11303.
- [102] J. Yang, L. Xiang, P. Chu, X. Wang, and C. Zhou, "Certified distributional robustness on smoothed classifiers," *IEEE Trans. Dependable Secure Comput.*, vol. 21, no. 2, pp. 876–888, Mar./Apr. 2024.
- [103] Z. Wang, Z. Zhou, and W. Liu, "DRF: Improving certified robustness via distributional robustness framework," in *Proc. AAAI Conf. Artif. Intell.*, 2024, vol. 38, pp. 15752–15760.
- [104] Z. Wang et al., "Towards transferable targeted adversarial examples," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 20534–20543.
- [105] P. Müller, A. Braun, and M. Keuper, "Classification robustness to common optical aberrations," in *Proc. Int. Conf. Comput. Vis.*, 2023, pp. 3632–3643.
- [106] E. Mintun, A. Kirillov, and S. Xie, "On interaction between augmentations and corruptions in natural corruption robustness," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 3571–3583.
- [107] D. Yin, R. Gontijo Lopes, J. Shlens, E. D. Cubuk, and J. Gilmer, "A fourier perspective on model robustness in computer vision," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 13276–13286.
- [108] Z. Liu et al., "AutoMix: Unveiling the power of mixup for stronger classifiers," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 441–458.
- [109] Z. Liu, Z. Zhao, and M. Larson, "Image shortcut squeezing: Countering perturbative availability poisons with compression," in *Proc. Int. Conf. Mach. Learn.*, 2023, pp. 22473–22487.
- [110] E. Borgnia, A. Ghiasi, J. Geiping, M. Goldblum, T. Goldstein, and A. Gupta, "Strong data augmentation sanitizes poisoning and backdoor attacks without an accuracy tradeoff," in *Proc. Int. Conf. Acoust., Speech Signal Process.*, 2021, pp. 3855–3859.
- [111] C. Xie, J. Wang, Z. Zhang, Z. Ren, and A. Yuille, "Mitigating adversarial effects through randomization," in *Proc. Int. Conf. Learn. Representations*, 2018.
- [112] J. Hayes, "On visible adversarial perturbations & digital watermarking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 1597–1604.
- [113] Y. Li, T. Zhai, Y. Jiang, Z. Li, and S.-T. Xia, "Backdoor attack in the physical world," in *Proc. Int. Conf. Learn. Representations Workshop*, 2021.
- [114] A. Levine and S. Feizi, "Robustness certificates for sparse adversarial attacks by randomized ablation," in *Proc. AAAI Conf. Artif. Intell.*, 2020, vol. 34, no. 04, pp. 4585–4593.
- [115] G.-H. Lee, Y. Yuan, S. Chang, and T. Jaakkola, "Tight certificates of adversarial robustness for randomly smoothed classifiers," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 4910–4921.
- [116] J. Jeong and J. Shin, "Multi-scale diffusion denoised smoothing," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 67374–67397.
- [117] A. Levine and S. Feizi, "Wasserstein smoothing: Certified robustness against wasserstein adversarial attacks," in *Proc. Int. Conf. Artif. Intell. Statist.*, 2020, pp. 3938–3947.
- [118] L. Li et al., "Tss: Transformation-specific smoothing for robustness certification," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, 2021, pp. 535–557.
- [119] S. Nandi, S. Addepalli, H. Rangwani, and R. V. Babu, "Certified adversarial robustness within multiple perturbation bounds," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 2297–2304.
- [120] B. Li, C. Chen, W. Wang, and L. Carin, "Certified adversarial robustness with additive noise," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 9464–9474.
- [121] J. Ren, Y. Zhou, J. Jin, L. Lyu, and D. Yan, "Dimension-independent certified neural network watermarks via mollifier smoothing," in *Proc. Int. Conf. Mach. Learn.*, 2023, pp. 28976–29008.
- [122] H. Salman, G. Yang, A. Kapoor, and J. Z. Kolter, "Denoised smoothing: A provable defense for pretrained classifiers," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 21945–21957.
- [123] J. Zhang, Z. Chen, H. Zhang, C. Xiao, and B. Li, "{DiffSmooth}: Certifiably robust learning via diffusion models and local smoothing," in *Proc. 32nd USENIX Secur.*, 2023, pp. 4787–4804.
- [124] Y. Scholten, J. Schuchardt, A. Bojchevski, and S. Günnemann, "Hierarchical randomized smoothing," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, 49783–49813.
- [125] Z. Hao, C. Ying, Y. Dong, H. Su, J. Song, and J. Zhu, "GSmooth: Certified robustness against semantic transformations via generalized randomized smoothing," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 8465–8483.
- [126] F. Stimberg et al., "Benchmarking robustness to adversarial image obfuscations," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 42830–42865.
- [127] E. D. Cubuk, B. Zoph, D. Mane, V. Vasudevan, and Q. V. Le, "AutoAugment: Learning augmentation policies from data," 2018, *arXiv:1805.09501*.
- [128] A. Athalye, N. Carlini, and D. Wagner, "Obfuscated gradients give a false sense of security: Circumventing defenses to adversarial examples," in *Proc. Int. Conf. Mach. Learn.*, 2018, pp. 274–283.
- [129] R. Zhai et al., "MACER: Attack-free and scalable robust training via maximizing certified radius," in *Proc. Int. Conf. Learn. Representations*, 2019.
- [130] A. Pal and J. Sulam, "Understanding noise-augmented training for randomized smoothing," *Trans. Mach. Learn. Res.*, 2023.
- [131] A. Kurakin, I. J. Goodfellow, and S. Bengio, "Adversarial examples in the physical world," in *Proc. Int. Conf. Learn. Representations Workshop*, 2016.
- [132] O. Poursaeed, I. Katsman, B. Gao, and S. Belongie, "Generative adversarial perturbations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 4422–4431.
- [133] Z. Huang and T. Zhang, "Black-box adversarial attack with transferable model-based embedding," in *Proc. Int. Conf. Learn. Representations*, 2019.
- [134] C.-C. Tu et al., "AutoZOOM: Autoencoder-based zeroth order optimization method for attacking black-box neural networks," in *Proc. AAAI Conf. Artif. Intell.*, 2019, vol. 33, pp. 742–749.
- [135] P.-Y. Chen, H. Zhang, Y. Sharma, J. Yi, and C.-J. Hsieh, "Zoo: Zeroth order optimization based black-box attacks to deep neural networks without training substitute models," in *Proc. 10th ACM Workshop Artif. Intell. Secur.*, 2017, pp. 15–26.
- [136] A. N. Bhagoji, W. He, B. Li, and D. Song, "Practical black-box attacks on deep neural networks using efficient query mechanisms," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 154–169.
- [137] A. Ilyas, L. Engstrom, and A. Madry, "Prior convictions: Black-box adversarial attacks with bandits and priors," in *Proc. Int. Conf. Learn. Representations*, 2019.
- [138] P. Zhao, P.-Y. Chen, S. Wang, and X. Lin, "Towards query-efficient black-box adversary with zeroth-order natural gradient descent," in *Proc. AAAI Conf. Artif. Intell.*, 2020, pp. 6909–6916.
- [139] J. Chen, M. I. Jordan, and M. J. Wainwright, "HopSkipJumpAttack: A query-efficient decision-based attack," in *Proc. IEEE Symp. Secur. Privacy*, 2020, pp. 1277–1294.
- [140] X. Wang et al., "Triangle attack: A query-efficient decision-based adversarial attack," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 156–174.
- [141] S. Cheng, Y. Dong, T. Pang, H. Su, and J. Zhu, "Improving black-box adversarial attacks with a transfer-based prior," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 10934–10944.
- [142] Z. Cai, C. Song, S. Krishnamurthy, A. Roy-Chowdhury, and S. Asif, "Blackbox attacks via surrogate ensemble search," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 5348–5362.
- [143] C. Ma, L. Chen, and J.-H. Yong, "Simulating unknown target models for query-efficient black-box attacks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 11830–11839.
- [144] M. Andriushchenko, F. Croce, N. Flammarion, and M. Hein, "Square attack: A query-efficient black-box adversarial attack via random search," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 484–501.
- [145] F. Croce, M. Andriushchenko, N. D. Singh, N. Flammarion, and M. Hein, "Sparse-RS: A versatile framework for query-efficient sparse black-box adversarial attacks," in *Proc. AAAI Conf. Artif. Intell.*, 2022, pp. 6437–6445.
- [146] Z. Chen, B. Li, S. Wu, S. Ding, and W. Zhang, "Query-efficient decision-based black-box patch attack," *IEEE Trans. Inf. Forensics Secur.*, vol. 18, pp. 5522–5536, 2023.
- [147] T. Brunner, F. Diehl, M. T. Le, and A. Knoll, "Guessing smart: Biased sampling for efficient black-box adversarial attacks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2019, pp. 4957–4965.
- [148] M. Shen, C. Li, H. Yu, Q. Li, L. Zhu, and K. Xu, "Decision-based query efficient adversarial attack via adaptive boundary learning," *IEEE Trans. Dependable Secure Comput.*, vol. 21, no. 4, pp. 1740–1753, Jul./Aug. 2024.
- [149] Y. Bai, Y. Zeng, Y. Jiang, S.-T. Xia, X. Ma, and Y. Wang, "Improving adversarial robustness via channel-wise activation suppressing," in *Proc. Int. Conf. Learn. Representations*, 2020.

- [150] S. Sarkar et al., "Robustness with query-efficient adversarial attack using reinforcement learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, 2023, pp. 2330–2337.
- [151] D. Su, H. Zhang, H. Chen, J. Yi, P.-Y. Chen, and Y. Gao, "Is robustness the cost of accuracy?—A comprehensive study on the robustness of 18 deep image classification models," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 631–648.
- [152] D. Wu, Y. Wang, S.-T. Xia, J. Bailey, and X. Ma, "Skip connections matter: On the transferability of adversarial examples generated with resnets," in *Proc. Int. Conf. Learn. Representations*, 2019.
- [153] Y. Zhang et al., "Why does little robustness help? A further step towards understanding adversarial transferability," in *Proc. IEEE Symp. Secur. Privacy*, 2023, pp. 3365–3384.
- [154] L. Wu and Z. Zhu, "Towards understanding and improving the transferability of adversarial examples in deep neural networks," in *Proc. 12th Asian Conf. Mach. Learn.*, 2020, pp. 837–850.
- [155] A. Ilyas, L. Engstrom, B. Tran, and A. Madry, "Adversarial examples are not bugs, they are features," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 125–136.
- [156] F. Tramèr, N. Papernot, I. Goodfellow, D. Boneh, and P. McDaniel, "The space of transferable adversarial examples," 2017, *arXiv:1704.03453*.
- [157] Y. Zhu et al., "Toward understanding and boosting adversarial transferability from a distribution perspective," *IEEE Trans. Image Process.*, vol. 31, pp. 6487–6501, 2022.
- [158] A. Demontis et al., "Why do adversarial attacks transfer? explaining transferability of evasion and poisoning attacks," in *Proc. 28th USENIX Secur.*, 2019, pp. 321–338.
- [159] C. Xie et al., "Improving transferability of adversarial examples with input diversity," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 2730–2739.
- [160] J. Zhang et al., "Improving the transferability of adversarial samples by path-augmented method," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 8173–8182.
- [161] K. Wang, X. He, W. Wang, and X. Wang, "Boosting adversarial transferability by block shuffle and rotation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 24336–24346.
- [162] S. Fang, J. Li, X. Lin, and R. Ji, "Learning to learn transferable attack," in *Proc. AAAI Conf. Artif. Intell.*, 2022, pp. 571–579.
- [163] A. Zhao, T. Chu, Y. Liu, W. Li, J. Li, and L. Duan, "Minimizing maximum model discrepancy for transferable black-box targeted attacks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 8153–8162.
- [164] X. Wang, J. Lin, H. Hu, J. Wang, and K. He, "Boosting adversarial transferability through enhanced momentum," in *Proc. Brit. Mach. Vis.*, 2021.
- [165] S. Long, W. Tao, L. Shuohao, J. Lei, and J. Zhang, "On the convergence of an adaptive momentum method for adversarial attacks," in *Proc. AAAI Conf. Artif. Intell.*, 2024, pp. 14132–14140.
- [166] J. Lin, C. Song, K. He, L. Wang, and J. E. Hopcroft, "Nesterov accelerated gradient and scale invariance for adversarial attacks," in *Proc. Int. Conf. Learn. Representations*, 2019.
- [167] Z. Xu, Z. Gu, J. Zhang, S. Cui, C. Meng, and W. Wang, "Backpropagation path search on adversarial transferability," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 4640–4650.
- [168] C. Gao et al., "Attacking transformers with feature diversity adversarial perturbation," in *Proc. AAAI Conf. Artif. Intell.*, 2024, pp. 1788–1796.
- [169] N. Xu, W. Feng, T. Zhang, and Y. Zhang, "A unified optimization framework for feature-based transferable attacks," *IEEE Trans. Inf. Forensics Secur.*, vol. 19, pp. 4794–4808, 2024.
- [170] K. Liang and B. Xiao, "StyLess: Boosting the transferability of adversarial examples," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 8163–8172.
- [171] R. Wang, Y. Guo, and Y. Wang, "AGS: Affordable and generalizable substitute training for transferable adversarial attack," in *Proc. AAAI Conf. Artif. Intell.*, 2024, vol. 38, pp. 5553–5562.
- [172] Z. Wei, J. Chen, Z. Wu, and Y. Jiang, "Adaptive cross-modal transferable adversarial attacks from images to videos," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 5, pp. 3772–3783, May 2024.
- [173] Z. Li, W. Wu, Y. Su, Z. Zheng, and M. R. Lyu, "CDTA: A cross-domain transfer-based attack with contrastive learning," in *Proc. AAAI Conf. Artif. Intell.*, 2023, vol. 37, pp. 1530–1538.
- [174] Z. Li, W. Wang, J. Li, K. Chen, and S. Zhang, "UCG: A universal cross-domain generator for transferable adversarial examples," *IEEE Trans. Inf. Forensics Secur.*, vol. 19, pp. 3023–3037, 2024.
- [175] H. Yang, J. Jeong, and K.-J. Yoon, "FACL-Attack: Frequency-aware contrastive learning for transferable adversarial attacks," in *Proc. AAAI Conf. Artif. Intell.*, 2024, pp. 6494–6502.
- [176] D. Lu, Z. Wang, T. Wang, W. Guan, H. Gao, and F. Zheng, "Set-level guidance attack: Boosting adversarial transferability of vision-language pre-training models," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 102–111.
- [177] P.-F. Zhang, Z. Huang, and G. Bai, "Universal adversarial perturbations for vision-language pre-trained models," in *Proc. 47th Int. ACM SIGIR Conf. Res. Develop. Inform. Retrieval*, 2024, pp. 862–871.
- [178] Z. Yin et al., "VLATTACK: Multimodal adversarial attacks on vision-language tasks via pre-trained models," in *Proc. Adv. Neural Inf. Process. Syst.*, 2023, pp. 52936–52956.
- [179] Y. Wang et al., "Generating transferable adversarial examples against vision transformers," in *Proc. 30th ACM Int. Conf. Multimedia*, 2022, pp. 5181–5190.
- [180] J. Zhang, Y. Huang, Z. Xu, W. Wu, and M. R. Lyu, "Improving the adversarial transferability of vision transformers with virtual dense connection," in *Proc. AAAI Conf. Artif. Intell.*, 2024, pp. 7133–7141.
- [181] J. Zhang, Y. Huang, W. Wu, and M. R. Lyu, "Transferable adversarial attacks on vision transformers with token gradient regularization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 16415–16424.
- [182] X. Wei and S. Zhao, "Boosting adversarial transferability with learnable patch-wise masks," *IEEE Trans. Multimedia*, vol. 26, pp. 3778–3787, 2024.
- [183] Q. Li, Y. Guo, and H. Chen, "Practical no-box adversarial attacks against DNNs," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 12849–12860.
- [184] F. Tramer, N. Carlini, W. Brendel, and A. Madry, "On adaptive attacks to adversarial example defenses," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 1633–1645.
- [185] M. Kang, D. Song, and B. Li, "DiffAttack: Evasion attacks against diffusion-based adversarial purification," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 73919–73942.
- [186] A. Shafahi, M. Najibi, Z. Xu, J. Dickerson, L. S. Davis, and T. Goldstein, "Universal adversarial training," in *Proc. AAAI Conf. Artif. Intell.*, 2020, pp. 5636–5643.
- [187] M. Li, C. Deng, T. Li, J. Yan, X. Gao, and H. Huang, "Towards transferable targeted attack," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 641–649.
- [188] Y. Dong et al., "Boosting adversarial attacks with momentum," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 9185–9193.
- [189] S. Jain and T. Dutta, "Towards understanding and improving adversarial robustness of vision transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 24736–24745.
- [190] M. Naseer, S. Khan, M. Hayat, F. S. Khan, and F. Porikli, "A self-supervised approach for adversarial robustness," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 259–268.
- [191] Z. Chen, B. Li, S. Wu, K. Jiang, S. Ding, and W. Zhang, "Content-based unrestricted adversarial attack," in *Proc. Adv. Neural Inf. Process. Syst.*, 2023, pp. 51719–51733.
- [192] X. Wang et al., "AdvST: Generating unrestricted adversarial images via style transfer," *IEEE Trans. Multimedia*, vol. 26, pp. 4846–4858, 2024.
- [193] S. Baluja and I. Fischer, "Learning to attack: Adversarial transformation networks," in *Proc. AAAI Conf. Artif. Intell.*, 2018, pp. 2687–2695.
- [194] C. Xiao, B. Li, J.-Y. Zhu, W. He, M. Liu, and D. Song, "Generating adversarial examples with adversarial networks," in *Proc. 27th Int. Joint Conf. Artif. Intell.*, 2018, pp. 3905–3911.
- [195] X. Jia, X. Wei, X. Cao, and X. Han, "Adv-watermark: A novel watermark perturbation for adversarial examples," in *Proc. 28th ACM Int. Conf. Multimedia*, 2020, pp. 1579–1587.
- [196] C. Zhang, K. Zhang, and Y. Li, "A causal view on robustness of neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 289–301.
- [197] D. Hendrycks, K. Zhao, S. Basart, J. Steinhardt, and D. Song, "Natural adversarial examples," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 15262–15271.
- [198] M. Arjovsky, L. Bottou, I. Gulrajani, and D. Lopez-Paz, "Invariant risk minimization," 2019, *arXiv:1907.02893*.
- [199] Y. Zhang et al., "Adversarial robustness through the lens of causality," in *Proc. Int. Conf. Learn. Representations*, 2021.
- [200] X.-C. Li, X.-Y. Zhang, F. Yin, and C.-L. Liu, "Decision-based adversarial attack with frequency mixup," *IEEE Trans. Inf. Forensics Secur.*, vol. 17, pp. 1038–1052, 2022.
- [201] Y. Dong, T. Pang, H. Su, and J. Zhu, "Evading defenses to transferable adversarial examples by translation-invariant attacks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 4307–4316.

- [202] A. Ilyas, L. Engstrom, A. Athalye, and J. Lin, "Black-box adversarial attacks with limited queries and information," in *Proc. Int. Conf. Mach. Learn.*, 2018, pp. 2137–2146.
- [203] Y. Li, L. Li, L. Wang, T. Zhang, and B. Gong, "NATTACK: Learning the distributions of adversarial examples for an improved black-box attack on deep neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 3866–3876.
- [204] H. Li, X. Xu, X. Zhang, S. Yang, and B. Li, "QEBA: Query-efficient boundary-based blackbox attack," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 1221–1230.
- [205] J. Huang, M. Wen, M. Wei, and Y. Bi, "Enhancing the transferability of adversarial samples with random noise techniques," *Comput. Secur.*, vol. 136, 2024, Art. no. 103541.
- [206] Y. Long et al., "Frequency domain model augmentation for adversarial attack," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 549–566.
- [207] Z. Li, W. Wang, J. Li, K. Chen, and S. Zhang, "Foolmix: Strengthen the transferability of adversarial examples by dual-blending and direction update strategy," *IEEE Trans. Inf. Forensics Secur.*, vol. 19, pp. 5286–5300, 2024.
- [208] Z. Yuan, J. Zhang, and S. Shan, "Adaptive image transformations for transfer-based adversarial attack," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 1–17.
- [209] P. Zhu, Z. Fan, S. Guo, K. Tang, and X. Li, "Improving adversarial transferability through hybrid augmentation," *Comput. Secur.*, vol. 139, 2024, Art. no. 103674.
- [210] X. Wang, Z. Zhang, and J. Zhang, "Structure invariant transformation for better adversarial transferability," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 4607–4619.
- [211] Z. Ge et al., "Improving the transferability of adversarial examples with arbitrary style transfer," in *Proc. 31st ACM Int. Conf. Multimedia*, 2023, pp. 4440–4449.
- [212] Y. Xiong, J. Lin, M. Zhang, J. E. Hopcroft, and K. He, "Stochastic variance reduced ensemble adversarial attack for boosting the adversarial transferability," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 14983–14992.
- [213] B. Chen, J. Yin, S. Chen, B. Chen, and X. Liu, "An adaptive model ensemble adversarial attack for boosting adversarial transferability," in *Proc. Int. Conf. Comput. Vis.*, 2023, pp. 4489–4498.
- [214] H. Chen, Y. Zhang, Y. Dong, X. Yang, H. Su, and J. Zhu, "Rethinking model ensemble in transfer-based adversarial attacks," in *Proc. Int. Conf. Learn. Representations*, 2023.
- [215] X. Wang and K. He, "Enhancing the transferability of adversarial attacks through variance tuning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 1924–1933.
- [216] Y. Guo, Q. Li, and H. Chen, "Backpropagating linearly improves transferability of adversarial examples," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 85–95.
- [217] N. Inkawhich, W. Wen, H. H. Li, and Y. Chen, "Feature space perturbations yield more transferable adversarial examples," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 7066–7074.
- [218] Q. Huang, I. Katsman, H. He, Z. Gu, S. Belongie, and S.-N. Lim, "Enhancing adversarial example transferability with an intermediate level attack," in *Proc. Int. Conf. Comput. Vis.*, 2019, pp. 4733–4742.
- [219] J. Zhang et al., "Improving adversarial transferability via neuron attribution-based attacks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 14993–15002.
- [220] W. Wu et al., "Boosting the transferability of adversarial samples via attention," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 1161–1170.
- [221] Z. Wang, H. Guo, Z. Zhang, W. Liu, Z. Qin, and K. Ren, "Feature importance-aware transferable adversarial attacks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 7639–7648.
- [222] Y. Wang and F. Farnia, "On the role of generalization in transferability of adversarial examples," in *Proc. 39th Uncertainty Artif. Intell.*, 2023, pp. 2259–2270.
- [223] Z. Wei, J. Chen, Z. Wu, and Y.-G. Jiang, "Cross-modal transferable adversarial attacks from images to videos," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 15044–15053.
- [224] K. Chen, Z. Wei, J. Chen, Z. Wu, and Y.-G. Jiang, "GCMA: Generative cross-modal transferable adversarial attacks from images to videos," in *Proc. 31st ACM Int. Conf. Multimedia*, 2023, pp. 698–708.
- [225] R. Wang, Y. Guo, and Y. Wang, "Global-local characteristic excited cross-modal attacks from images to videos," in *Proc. AAAI Conf. Artif. Intell.*, 2023, vol. 37, pp. 2635–2643.
- [226] M. Salzmann et al., "Learning transferable adversarial perturbations," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 13950–13962.
- [227] Y. Lu et al., "Enhancing cross-task black-box transferability of adversarial examples with dispersion reduction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 940–949.
- [228] W. Feng, N. Xu, T. Zhang, Y. Zhang, and F. Wu, "Enhancing cross-task transferability of adversarial examples via spatial and channel attention," *IEEE Trans. Multimedia*, vol. 28, pp. 2016–2030, 2026.
- [229] H. Luo, J. Gu, F. Liu, and P. Torr, "An image is worth 1000 lies: Transferability of adversarial images across prompts on vision-language models," in *Proc. Int. Conf. Learn. Representations*, 2023.
- [230] Y. Ban and Y. Dong, "Pre-trained adversarial perturbations," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 1196–1209.
- [231] Z. Zhou et al., "Downstream-agnostic adversarial examples," in *IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 4322–4332.
- [232] M. Naseer, K. Ranasinghe, S. Khan, F. Khan, and F. Porikli, "On improving adversarial transferability of vision transformers," in *Proc. Int. Conf. Learn. Representations*, 2022.
- [233] K. Mahmood, R. Mahmood, and M. Van Dijk, "On the robustness of vision transformers to adversarial examples," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 7818–7827.
- [234] H. Wang, X. Wu, Z. Huang, and E. P. Xing, "High-frequency component helps explain the generalization of convolutional neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 8684–8694.
- [235] Y. Bai, J. Mei, A. L. Yuille, and C. Xie, "Are transformers more robust than CNNs?," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 26831–26843.
- [236] R. Shao, Z. Shi, J. Yi, P.-Y. Chen, and C.-j. Hsieh, "On the adversarial robustness of vision transformers," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022.
- [237] Z. Chen, C. Xu, H. Lv, S. Liu, and Y. Ji, "Understanding and improving adversarial transferability of vision transformers and convolutional neural networks," *Inf. Sci.*, vol. 648, 2023, Art. no. 119474.
- [238] W. Ma, Y. Li, X. Jia, and W. Xu, "Transferable adversarial attack for both vision transformers and convolutional networks via momentum integrated gradients," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 4630–4639.
- [239] B. Tang, Z. Wang, Y. Bin, Q. Dou, Y. Yang, and H. T. Shen, "Ensemble diversity facilitates adversarial transferability," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 24377–24386.
- [240] F. Tramèr, A. Kurakin, N. Papernot, I. Goodfellow, D. Boneh, and P. McDaniel, "Ensemble adversarial training: Attacks and defenses," in *Proc. Int. Conf. Learn. Representations*, 2017.
- [241] G. W. Ding, Y. Sharma, Lui, and R. Huang, "MMA training: Direct input space margin maximization through adversarial training," in *Proc. Int. Conf. Learn. Representations*, 2018.
- [242] A. Ghiasi, A. Shafahi, and T. Goldstein, "Breaking certified defenses: Semantic adversarial examples with spoofed robustness certificates," in *Proc. Int. Conf. Learn. Representations*, 2019.
- [243] F. Croce and M. Hein, "Reliable evaluation of adversarial robustness with an ensemble of diverse parameter-free attacks," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 2206–2216.
- [244] J. Rony, L. G. Hafemann, L. S. Oliveira, I. B. Ayed, R. Sabourin, and E. Granger, "Decoupling direction and norm for efficient gradient-based L2 adversarial attacks and defenses," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 4317–4325.
- [245] Y. Yu and C.-Z. Xu, "Efficient loss function by minimizing the detrimental effect of floating-point errors on gradient-based attacks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 4056–4066.
- [246] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, "The unreasonable effectiveness of deep features as a perceptual metric," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 586–595.
- [247] F. Croce and M. Hein, "Minimally distorted adversarial examples with a fast adaptive boundary attack," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 2196–2205.
- [248] W. Brendel, J. Rauber, M. Kümmeler, I. Ustyuzhaninov, and M. Bethge, "Accurate, Reliable and Fast Robustness Evaluation," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 12817–12827.
- [249] M. Pintor, F. Roli, W. Brendel, and B. Biggio, "Fast minimum-norm adversarial attacks through adaptive norm constraints," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 20052–20062.

- [250] Y. Dong et al., "How robust is google's bard to adversarial image attacks?," 2023, *arXiv:2309.11751*.
- [251] Y. Zhao et al., "On evaluating adversarial robustness of large vision-language models," 2023, *arXiv:2305.16934*.
- [252] Y. Wang, W. Hu, Y. Dong, J. Liu, H. Zhang, and R. Hong, "Align is not enough: Multimodal universal jailbreak attack against multimodal large language models," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 6, pp. 5475–5488, Jun. 2025.
- [253] Y. Gong et al., "Figstep: Jailbreaking large vision-language models via typographic visual prompts," in *Proc. AAAI Conf. Artif. Intell.*, 2025, pp. 23951–23959.
- [254] Y. Li, H. Guo, K. Zhou, W. X. Zhao, and J.-R. Wen, "Images are achilles' heel of alignment: Exploiting visual vulnerabilities for jailbreaking multimodal large language models," in *Proc. Eur. Conf. Comput. Vis.*, 2024, pp. 174–189.
- [255] Z. Yang et al., "Distraction is all you need for multimodal large language model jailbreaking," in *Proc. IEEE/CVF Comput. Vis. Pattern Recognit. Conf.*, 2025, pp. 9467–9476.
- [256] E. Shayegani, Y. Dong, and N. Abu-Ghazaleh, "Jailbreak in pieces: Compositional adversarial attacks on multi-modal language models," in *Proc. Int. Conf. Learn. Representations*, 2023.
- [257] W. R. Huang, J. Geiping, L. Fowl, G. Taylor, and T. Goldstein, "MetaPoison: Practical general-purpose clean-label data poisoning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 12080–12091.
- [258] A. Turner, D. Tsipras, and A. Madry, "Label-consistent backdoor attacks," 2019, *arXiv:1912.02771*.
- [259] Y. Gao, Y. Li, L. Zhu, D. Wu, Y. Jiang, and S.-T. Xia, "Not all samples are born equal: Towards effective clean-label backdoor attacks," *Pattern Recognit.*, vol. 139, 2023, Art. no. 109512.
- [260] Y. Zeng, M. Pan, H. A. Just, L. Lyu, M. Qiu, and R. Jia, "Narcissus: A practical clean-label backdoor attack with limited information," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, 2023, pp. 771–785.
- [261] Y. Li, Y. Li, B. Wu, L. Li, R. He, and S. Lyu, "Invisible backdoor attack with sample-specific triggers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 16463–16472.
- [262] T. A. Nguyen and A. Tran, "Input-aware dynamic backdoor attack," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 3454–3464.
- [263] S. Li, M. Xue, B. Z. H. Zhao, H. Zhu, and X. Zhang, "Invisible backdoor attacks on deep neural networks via steganography and regularization," *IEEE Trans. Dependable Secure Comput.*, vol. 18, no. 5, pp. 2088–2105, Sep./Oct. 2021.
- [264] Y. Zeng, W. Park, Z. M. Mao, and R. Jia, "Rethinking the backdoor attacks' triggers: A frequency perspective," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 16453–16461.
- [265] H. Huang, X. Ma, S. M. Erfani, J. Bailey, and Y. Wang, "Unlearnable examples: Making personal data unexploitable," in *Proc. Int. Conf. Learn. Representations*, 2020.
- [266] S. Fu, F. He, Y. Liu, L. Shen, and D. Tao, "Robust unlearnable examples: Protecting data privacy against adversarial learning," in *Proc. Int. Conf. Learn. Representations*, 2021.
- [267] C. Zhu, W. R. Huang, H. Li, G. Taylor, C. Studer, and T. Goldstein, "Transferable clean-label poisoning attacks on deep neural nets," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 7614–7623.
- [268] H. Aghakhani, D. Meng, Y.-X. Wang, C. Kruegel, and G. Vigna, "Bulls-eye polytope: A scalable clean-label poisoning attack with improved transferability," in *Proc. IEEE Eur. Symp. Secur. Privacy*, 2021, pp. 159–178.
- [269] J. Geiping et al., "Witches' brew: Industrial scale data poisoning via gradient matching," in *Proc. Int. Conf. Learn. Representations*, 2020.
- [270] J. Zhang et al., "Unlearnable clusters: Towards label-agnostic unlearnable examples," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 3984–3993.
- [271] P. He et al., "Sharpness-aware data poisoning attack," in *Proc. Int. Conf. Learn. Representations*, 2024.
- [272] J. Ren, H. Xu, Y. Wan, X. Ma, L. Sun, and J. Tang, "Transferable unlearnable examples," in *Proc. Int. Conf. Learn. Representations*, 2023.
- [273] H. Liu, J. Jia, and N. Z. Gong, "{PoisonedEncoder}: Poisoning the unlabeled pre-training data in contrastive learning," in *31st USENIX Secur. Symp.*, 2022, pp. 3629–3645.
- [274] A. Saha, A. Tejankar, S. A. Koohpayegani, and H. Pirsiavash, "Backdoor attacks on self-supervised learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 13327–13336.
- [275] J. Jia, Y. Liu, and N. Z. Gong, "BadEncoder: Backdoor attacks to pre-trained encoders in self-supervised learning," in *Proc. IEEE Symp. Secur. Privacy*, 2022, pp. 2043–2059.
- [276] C. Li et al., "An embarrassingly simple backdoor attack on self-supervised learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 4367–4378.
- [277] G. Tao, Z. Wang, S. Feng, G. Shen, S. Ma, and X. Zhang, "Distribution preserving backdoor attack in self-supervised learning," in *Proc. IEEE Symp. Secur. Privacy*, 2024, pp. 2029–2047.
- [278] J. Zhang, H. Liu, J. Jia, and N. Z. Gong, "Data poisoning based backdoor attacks to contrastive learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 24357–24366.
- [279] H. He, K. Zha, and D. Katabi, "Indiscriminate poisoning attacks on unsupervised contrastive learning," in *Proc. Int. Conf. Learn. Representations*, 2022.
- [280] Y. Liu, X. Ma, J. Bailey, and F. Lu, "Reflection backdoor: A natural backdoor attack on deep neural networks," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 182–199.
- [281] H. Zhong, C. Liao, A. C. Squicciarini, S. Zhu, and D. Miller, "Backdoor embedding in convolutional neural network models via invisible perturbation," in *Proc. 10th ACM Conf. Data Application Secur. Privacy*, 2020, pp. 97–108.
- [282] A. Saha, A. Subramanya, and H. Pirsiavash, "Hidden trigger backdoor attacks," in *Proc. AAAI Conf. Artif. Intell.*, 2020, pp. 11957–11965.
- [283] H. Sourli, L. Fowl, R. Chellappa, M. Goldblum, and T. Goldstein, "Sleeping agent: Scalable hidden trigger backdoors for neural networks trained from scratch," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 19165–19178.
- [284] A. Saleem, R. Wen, M. Backes, S. Ma, and Y. Zhang, "Dynamic backdoor attacks against machine learning models," in *Proc. IEEE 7th Eur. Symp. Secur. Privacy*, 2022, pp. 703–718.
- [285] E. Quiring and K. Rieck, "Backdooring and poisoning neural networks with image-scaling attacks," in *Proc. IEEE Secur. Privacy Workshops*, 2020, pp. 41–47.
- [286] B. Fang, B. Li, S. Wu, S. Ding, R. Yi, and L. Ma, "Re-thinking data availability attacks against deep neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 12215–12224.
- [287] T. A. Nguyen and A. T. Tran, "Wanet-imperceptible warping-based backdoor attack," in *Proc. Int. Conf. Learn. Representations*, 2020.
- [288] R. Wang, R. Wan, Z. Guo, Q. Guo, and R. Huang, "SPY-watermark: Robust invisible watermarking for backdoor attack," in *Proc. Int. Conf. Acoust., Speech Signal Process.*, 2024, pp. 2700–2704.
- [289] C. Xu, W. Liu, Y. Zheng, S. Wang, and C.-H. Chang, "An imperceptible data augmentation based blackbox clean-label backdoor attack on deep neural networks," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 70, no. 12, pp. 5011–5024, Dec. 2023.
- [290] S. Cheng, Y. Liu, S. Ma, and X. Zhang, "Deep feature space trojan attack of neural networks by controlled detoxification," in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, pp. 1148–1156.
- [291] Y. Gao et al., "A dual stealthy backdoor: From both spatial and frequency perspectives," in *Proc. AAAI Conf. Artif. Intell.*, 2024, pp. 1851–1859.
- [292] L. Tao, L. Feng, J. Yi, S.-J. Huang, and S. Chen, "Better safe than sorry: Preventing delusive adversaries with adversarial training," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 16209–16225.
- [293] B. Wang et al., "Neural cleanse: Identifying and mitigating backdoor attacks in neural networks," in *Proc. IEEE Symp. Secur. Privacy*, 2019, pp. 707–723.
- [294] L. Tao, L. Feng, H. Wei, J. Yi, S.-J. Huang, and S. Chen, "Can adversarial training be manipulated by non-robust features?," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 26504–26518.
- [295] A. Subramanya, S. A. Koohpayegani, A. Saha, A. Tejankar, and H. Pirsiavash, "A closer look at robustness of vision transformers to backdoor attacks," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis.*, 2024, pp. 3874–3883.
- [296] K. D. Doan, Y. Lao, P. Yang, and P. Li, "Defending backdoor attacks on vision transformer via patch processing," in *Proc. AAAI Conf. Artif. Intell.*, 2023, pp. 506–515.
- [297] J. Yoon, S. J. Hwang, and J. Lee, "Adversarial purification with score-based generative models," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 12062–12072.
- [298] W. Nie, B. Guo, Y. Huang, C. Xiao, A. Vahdat, and A. Anandkumar, "Diffusion models for adversarial purification," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 16805–16827.
- [299] K. Song, H. Lai, Y. Pan, and J. Yin, "MimicDiffusion: Purifying adversarial perturbation via mimicking clean diffusion model," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 24665–24674.
- [300] M. Andriushchenko and N. Flammarion, "Understanding and improving fast adversarial training," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 16048–16059.

- [301] H. Kim, W. Lee, and J. Lee, "Understanding catastrophic overfitting in single-step adversarial training," in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 8119–8127.
- [302] A. Shafahi et al., "Adversarial training for free!," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 3358–3369.
- [303] E. Wong, L. Rice, and J. Z. Kolter, "Fast is better than free: Revisiting adversarial training," in *Proc. Int. Conf. Learn. Representations*, 2019.
- [304] X. Jia, Y. Zhang, B. Wu, J. Wang, and X. Cao, "Boosting fast adversarial training with learnable adversarial initialization," *IEEE Trans. Image Process.*, vol. 31, pp. 4417–4430, 2022.
- [305] X. Jia et al., "Improving fast adversarial training with prior-guided knowledge," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 9, pp. 6367–6383, Sep. 2024.
- [306] R. Lin, C. Yu, and T. Liu, "Eliminating catastrophic overfitting via abnormal adversarial examples regularization," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 67866–67885.
- [307] J. Dong, S.-M. Moosavi-Dezfooli, J. Lai, and X. Xie, "The enemy of my enemy is my friend: Exploring inverse adversaries for improving adversarial training," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 24678–24687.
- [308] K. Tong, C. Jiang, J. Gui, and Y. Cao, "Taxonomy driven fast adversarial training," in *Proc. AAAI Conf. Artif. Intell.*, 2024, pp. 5233–5242.
- [309] G. Y. Park and S. W. Lee, "Reliably fast adversarial training via latent adversarial perturbation," in *Proc. Int. Conf. Comput. Vis.*, 2021, pp. 7758–7767.
- [310] B. Vivek and R. V. Babu, "Single-step adversarial training with dropout scheduling," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 947–956.
- [311] Z. He, T. Li, S. Chen, and X. Huang, "Investigating catastrophic overfitting in fast adversarial training: A self-fitting perspective," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 2313–2320.
- [312] G. Ortiz-Jimenez et al., "Catastrophic overfitting can be induced with discriminative non-robust features," *Trans. Mach. Learn. Res.*.
- [313] R. Lin, C. Yu, B. Han, and T. Liu, "On the over-memorization during natural, robust and catastrophic overfitting," in *Proc. Int. Conf. Learn. Representations*, 2023.
- [314] Z. Huang et al., "Fast adversarial training with adaptive step size," *IEEE Trans. Image Process.*, vol. 32, pp. 6102–6114, 2023.
- [315] E. A. Rocamora, F. Liu, G. Chrysos, P. M. Olmos, and V. Cevher, "Efficient local linearity regularization to overcome catastrophic overfitting," in *Proc. Int. Conf. Learn. Representations*, 2023.
- [316] Z. Li, D. Yu, M. Wu, S. Chan, H. Yu, and Z. Han, "Revisiting single-step adversarial training for robustness and generalization," *Pattern Recognit.*, vol. 151, 2024, Art. no. 110356.
- [317] M. Zhao, L. Zhang, Y. Kong, and B. Yin, "Fast adversarial training with smooth convergence," in *Proc. Int. Conf. Comput. Vis.*, 2023, pp. 4720–4729.
- [318] P. de Jorge Aranda et al., "Make some noise: Reliable and efficient single-step adversarial training," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 12881–12893.
- [319] T. Li, Y. Wu, S. Chen, K. Fang, and X. Huang, "Subspace adversarial training," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 13409–13418.
- [320] J. Grabinski, S. Jung, J. Keuper, and M. Keuper, "Frequencylowcut pooling-plugin and play against catastrophic overfitting," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 36–57.
- [321] S. Lee, H. Lee, and S. Yoon, "Adversarial vertex mixup: Toward better adversarially robust generalization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 272–281.
- [322] C. Song et al., "Regional adversarial training for better robust generalization," *Int. J. Comput. Vis.*, vol. 132, pp. 4510–4520, 2024.
- [323] J. Zhang, J. Zhu, G. Niu, B. Han, M. Sugiyama, and M. Kankanhalli, "Geometry-aware instance-reweighted adversarial training," in *Proc. Int. Conf. Learn. Representations*, 2020.
- [324] M. Cheng, Q. Lei, P. Y. Chen, I. Dhillon, and C. J. Hsieh, "Cat: Customized adversarial training for improved robustness," in *Proc. Int. Joint Conf. Artif. Intell.*, 2022, pp. 673–679.
- [325] C. Yu et al., "Understanding robust overfitting of adversarial training and beyond," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 25595–25610.
- [326] C. Song, K. He, J. Lin, L. Wang, and J. E. Hoyer, "Robust local features for improving the generalization of adversarial training," in *Proc. Int. Conf. Learn. Representations*, 2019.
- [327] D. Stutz, M. Hein, and B. Schiele, "Confidence-calibrated adversarial training: Generalizing to unseen attacks," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 9155–9166.
- [328] Y. Dong et al., "Exploring memorization in adversarial training," in *Proc. Int. Conf. Learn. Representations*, 2021.
- [329] H. Liu, Z. Zhong, N. Sebe, and S. Satoh, "Mitigating robust overfitting via self-residual-calibration regularization," *Artif. Intell.*, vol. 317, 2023, Art. no. 103877.
- [330] A. Liu et al., "Exploring the relationship between architectural design and adversarially robust generalization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 4096–4107.
- [331] P. Izmailov, D. Podoprikin, T. Garipov, D. Vetrov, and A. G. Wilson, "Averaging weights leads to wider optima and better generalization," in *Proc. 34th Conf. Uncertainty Artif. Intell.*, 2018, pp. 876–885.
- [332] T. Chen, Z. Zhang, S. Liu, S. Chang, and Z. Wang, "Robust overfitting may be mitigated by properly learned smoothening," in *Proc. Int. Conf. Learn. Representations*, 2021.
- [333] S.-A. Rebuffi, S. Goyal, D. A. Calian, F. Stimberg, O. Wiles, and T. Mann, "Fixing data augmentation to improve adversarial robustness," 2021, *arXiv:2103.01946*.
- [334] D. Wu, S.-T. Xia, and Y. Wang, "Adversarial weight perturbation helps robust generalization," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 2958–2969.
- [335] L. Li and M. Spratling, "Understanding and combating robust overfitting via input loss landscape analysis and regularization," *Pattern Recognit.*, vol. 136, 2023, Art. no. 109229.
- [336] J.-B. Alayrac, J. Uesato, P.-S. Huang, A. Fawzi, R. Stanforth, and P. Kohli, "Are labels required for improving adversarial robustness?," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 12214–12223.
- [337] C. P. Lau, J. Liu, H. Soury, W.-A. Lin, S. Feizi, and R. Chellappa, "Interpolated joint space adversarial training for robust and generalizable defenses," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 11, pp. 13054–13067, Nov. 2023.
- [338] Y. Wang, L. Li, J. Yang, Z. Lin, and Y. Wang, "Balance, imbalance, and rebalance: Understanding robust overfitting from a min-max game perspective," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 15775–15798.
- [339] V. Singla, S. Singla, S. Feizi, and D. Jacobs, "Low curvature activations reduce overfitting in adversarial training," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 16403–16413.
- [340] C. Xie and A. Yuille, "Intriguing properties of adversarial training at scale," in *Proc. Int. Conf. Learn. Representations*, 2019.
- [341] P. Hou, J. Han, and X. Li, "Improving adversarial robustness with self-paced hard-class pair reweighting," in *Proc. AAAI Conf. Artif. Intell.*, 2023, pp. 14883–14891.
- [342] L. He, Q. Ai, Y. Lei, L. Pan, Y. Ren, and Z. Xu, "Edge enhancement improves adversarial robustness in image classification," *Neurocomputing*, vol. 518, pp. 122–132, 2023.
- [343] P. Nakkiran, "Adversarial robustness may be at odds with simplicity," 2019, *arXiv:1901.00532*.
- [344] C. Xie, Y. Wu, L. v. d. Maaten, A. L. Yuille, and K. He, "Feature denoising for improving adversarial robustness," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 501–509.
- [345] W. J. Kim, Y. Cho, J. Jung, and S.-E. Yoon, "Feature separation and recalibration for adversarial robustness," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 8183–8192.
- [346] G. Jin, X. Yi, W. Huang, S. Schewe, and X. Huang, "Enhancing adversarial training with second-order statistics of weights," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 15273–15283.
- [347] H. Wang, Y. Deng, S. Yoo, and Y. Lin, "Exploring robust features for improving adversarial robustness," *IEEE Trans. Cybern.*, vol. 54, no. 9, pp. 5141–5151, Sep. 2024.
- [348] C. Mao, Z. Zhong, J. Yang, C. Vondrick, and B. Ray, "Metric learning for adversarial robustness," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 480–491.
- [349] H. Kuang, H. Liu, X. Lin, and R. Ji, "Defense against adversarial attacks using topology aligning adversarial training," *IEEE Trans. Inf. Forensics Secur.*, vol. 19, pp. 3659–3673, 2024.
- [350] H. Kuang, H. Liu, Y. Wu, S. Satoh, and R. Ji, "Improving adversarial robustness via information bottleneck distillation," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 10796–10813.
- [351] P. Awasthi, A. Mao, M. Mohri, and Y. Zhong, "Theoretically grounded loss functions and algorithms for adversarial robustness," in *Proc. Int. Conf. Artif. Intell. Statist.*, 2023, pp. 10077–10094.
- [352] M. Kim, J. Tack, and S. J. Hwang, "Adversarial self-supervised contrastive learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 2983–2994.

- [353] H. Xu, X. Liu, Y. Li, A. Jain, and J. Tang, "To be robust or to be fair: Towards fairness in adversarial training," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 11492–11501.
- [354] C. Sun, C. Xu, Y. Wu, D. Liang, X. Liu, and A. Liu, "Improving robust fairness via balance adversarial training," in *Proc. AAAI Conf. Artif. Intell.*, 2023, vol. 37, pp. 15161–15169.
- [355] Z. Wei, Y. Wang, Y. Guo, and Y. Wang, "Cfa: Class-wise calibrated fair adversarial training," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 8193–8201.
- [356] P. Benz, C. Zhang, A. Karjauv, and I. S. Kweon, "Robustness may be at odds with fairness: An empirical study on class-wise accuracy," in *Proc. Adv. Neural Inf. Process. Syst. Pre-Registration Workshop*, 2021, pp. 325–342.
- [357] X. Zhou, N. Yang, and O. Wu, "Combining adversaries with anti-adversaries in training," in *Proc. AAAI Conf. Artif. Intell.*, 2023, pp. 11435–11442.
- [358] R. Rade and S.-M. Moosavi-Dezfooli, "Reducing excessive margin to achieve a better accuracy vs. robustness trade-off," in *Proc. Int. Conf. Learn. Representations*, 2021.
- [359] J.-L. Yin, B. Chen, W. Zhu, B.-H. Chen, and X. Liu, "Push stricter to decide better: A class-conditional feature adaptive framework for improving adversarial robustness," *IEEE Trans. Inf. Forensics Secur.*, vol. 18, pp. 2119–2131, 2023.
- [360] G. Li, W. Tong, and T. Yang, "Maximization of average precision for deep learning with adversarial ranking robustness," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 15475–15496.
- [361] J. Yan, H. Yin, Z. Zhao, W. Ge, and J. Zhang, "Enhance adversarial robustness via geodesic distance," *IEEE Trans. Artif. Intell.*, vol. 5, no. 8, pp. 4202–4216, Aug. 2024.
- [362] X. Yu et al., "Improving adversarial robustness by learning shared information," *Pattern Recognit.*, vol. 134, 2023, Art. no. 109054.
- [363] H. Zhang and J. Wang, "Defense against adversarial attacks using feature scattering-based adversarial training," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 1831–1841.
- [364] J. Wang and H. Zhang, "Bilateral adversarial training: towards fast training of more robust models against adversarial attacks," in *Proc. Int. Conf. Comput. Vis.*, 2019, pp. 6629–6638.
- [365] C. Shao, W. Li, J. Huo, Z. Feng, and Y. Gao, "Attention-based investigation and solution to the trade-off issue of adversarial training," *Neural Netw.*, vol. 174, 2024, Art. no. 106224.
- [366] Y. Mao, M. Müller, M. Fischer, and M. Vechev, "Connecting certified and adversarial training," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 73422–73440.
- [367] S. Gowal, S.-A. Rebuffi, O. Wiles, F. Stimberg, D. A. Calian, and T. A. Mann, "Improving robustness using generated data," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 4218–4233.
- [368] S. Kanai, S. Yamaguchi, M. Yamada, H. Takahashi, K. Ohno, and Y. Ida, "One-vs-the-rest loss to focus on important samples in adversarial training," in *Proc. Int. Conf. Mach. Learn.*, 2023, pp. 15669–15695.
- [369] J. Zhang et al., "Attacks which do not kill training make adversarial learning stronger," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 11278–11287.
- [370] M. Dong and C. Xu, "Adversarial robustness via random projection filters," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 4077–4086.
- [371] X. Wei, S. Zhao, and B. Li, "Revisiting the trade-off between accuracy and robustness via weight distribution of filters," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 8870–8882, Dec. 2024.
- [372] G. Jin, X. Yi, D. Wu, R. Mu, and X. Huang, "Randomized adversarial training via taylor expansion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 16447–16457.
- [373] S. Kundu, S. Sundaresan, M. Pedram, and P. A. Beerel, "FLOAT: Fast learnable once-for-all adversarial training for tunable trade-off between accuracy and robustness," in *Proc. Winter Conf. Appl. Comput. Vis.*, 2023, pp. 2349–2358.
- [374] K. Zhu, X. Hu, J. Wang, X. Xie, and G. Yang, "Improving generalization of adversarial training via robust critical fine-tuning," in *Proc. Int. Conf. Comput. Vis.*, 2023, pp. 4424–4434.
- [375] S. Gowda, B. Zonooz, and E. Arani, "Conserve-update-revise to cure generalization and robustness trade-off in adversarial training," in *Proc. Int. Conf. Learn. Representations*, 2014.
- [376] C. Xie, M. Tan, B. Gong, J. Wang, A. L. Yuille, and Q. V. Le, "Adversarial examples improve image recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 816–825.
- [377] H. Wang, A. Zhang, S. Zheng, X. Shi, M. Li, and Z. Wang, "Removing batch normalization boosts adversarial training," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 23433–23445.
- [378] M. Naseer, S. Khan, and F. Porikli, "Local gradients smoothing: Defense against localized adversarial attacks," in *Proc. Winter Conf. Appl. Comput. Vis.*, 2019, pp. 1300–1307.
- [379] K. Xu, Y. Xiao, Z. Zheng, K. Cai, and R. Nevatia, "PatchZero: Defending against adversarial patch attacks by detecting and zeroing the patch," in *Proc. Winter Conf. Appl. Comput. Vis.*, 2023, pp. 4632–4641.
- [380] F. Tramèr and D. Boneh, "Adversarial training and robustness for multiple perturbations," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 5866–5876.
- [381] P. Maini, E. Wong, and Z. Kolter, "Adversarial robustness against the union of multiple perturbation models," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 6640–6650.
- [382] L. Hsiung, Y.-Y. Tsai, P.-Y. Chen, and T.-Y. Ho, "Towards compositional adversarial robustness: Generalizing adversarial training to composite semantic perturbations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 24658–24667.
- [383] S. Dai, S. Mahloujifar, and P. Mittal, "Formulating robustness against unforeseen attacks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 8647–8661.
- [384] Z. Jiang, T. Chen, T. Chen, and Z. Wang, "Robust pre-training by adversarial contrastive learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 16199–16210.
- [385] Z. Li, D. Yu, M. Wu, C. Jin, and H. Yu, "Adversarial supervised contrastive learning," *Mach. Learn.*, vol. 112, no. 6, pp. 2105–2130, 2023.
- [386] D. Hendrycks, K. Lee, and M. Mazeika, "Using pre-training can improve model robustness and uncertainty," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 2712–2721.
- [387] Z. Liu, Y. Xu, X. Ji, and A. B. Chan, "TWINS: A fine-tuning framework for improved transferability of adversarial robustness and generalization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 16436–16446.
- [388] X. Xu, J. Zhang, and M. Kankanhalli, "AutoLoRa: A parameter-free automated robust fine-tuning framework," in *Proc. Int. Conf. Learn. Representations*, 2023.
- [389] Z. Zhou et al., "Securely fine-tuning pre-trained encoders against adversarial examples," in *Proc. IEEE Symp. Secur. Privacy*, 2024, pp. 3015–3033.
- [390] D. Zhang, T. Zhang, Y. Lu, Z. Zhu, and B. Dong, "You only propagate once: Accelerating adversarial training via maximal principle," in *Proc. Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 227–238.
- [391] W. Hua, Y. Zhang, C. Guo, Z. Zhang, and G. E. Suh, "BulletTrain: Accelerating robust neural network training via boundary example mining," in *Proc. Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 18527–18538.
- [392] X. Jia, J. Li, J. Gu, Y. Bai, and X. Cao, "Fast propagation is better: Accelerating single-step adversarial training via sampling subnetworks," *IEEE Trans. Inf. Forensics Secur.*, vol. 19, pp. 4547–4559, 2024.
- [393] L. Rice, E. Wong, and Z. Kolter, "Overfitting in adversarially robust deep learning," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 8093–8104.
- [394] S.-A. Rebuffi, S. Gowal, D. A. Calian, F. Stimberg, O. Wiles, and T. A. Mann, "Data augmentation can improve robustness," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 29935–29948.
- [395] J. Tack, S. Yu, J. Jeong, M. Kim, S. J. Hwang, and J. Shin, "Consistency regularization for adversarial robustness," in *Proc. Proc. AAAI Conf. Artif. Intell.*, 2022, pp. 8414–8422.
- [396] L. Li and M. W. Spratling, "Data augmentation alone can improve adversarial training," in *Proc. Int. Conf. Learn. Representations*, 2022.
- [397] Q.-Z. Cai, C. Liu, and D. Song, "Curriculum adversarial training," in *Proc. Int. Joint Conf. Artif. Intell.*, 2018, pp. 3740–3747.
- [398] Y. Wang, X. Ma, J. Bailey, J. Yi, B. Zhou, and Q. Gu, "On the convergence and robustness of adversarial training," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 6586–6595.
- [399] T. Pang, K. Xu, C. Du, N. Chen, and J. Zhu, "Improving adversarial robustness via promoting ensemble diversity," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 4970–4979.
- [400] D. Tsipras, S. Santurkar, L. Engstrom, A. Turner, and A. Madry, "Robustness may be at odds with accuracy," in *Proc. Int. Conf. Learn. Representations*, 2018.
- [401] C. Xie, M. Tan, B. Gong, A. Yuille, and Q. V. Le, "Smooth adversarial training," 2020, *arXiv:2006.14536*.
- [402] J. Cui, S. Liu, L. Wang, and J. Jia, "Learnable boundary guided adversarial training," in *Proc. Int. Conf. Comput. Vis.*, 2021, pp. 15721–15730.

- [403] H. Wang and Y. Wang, "Generalist: Decoupling natural and robust generalization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 20554–20563.
- [404] Y.-Y. Yang, C. Rashtchian, H. Zhang, and K. Chaudhuri, "A closer look at accuracy vs. robustness," in *Proc. Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 8588–8601.
- [405] A. Raghunathan, S. M. Xie, F. Yang, J. C. Duchi, and P. Liang, "Understanding and mitigating the tradeoff between robustness and accuracy," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 7909–7919.
- [406] T. Chen, S. Liu, S. Chang, Y. Cheng, L. Amini, and Z. Wang, "Adversarial robustness: From self-supervised pre-training to fine-tuning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 699–708.
- [407] B. Wu, J. Gu, Z. Li, D. Cai, X. He, and W. Liu, "Towards efficient adversarial training on vision transformers," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 307–325.
- [408] E. Debenedetti, V. Sehwag, and P. Mittal, "A light recipe to train robust vision transformers," in *Proc. IEEE Conf. Secure Trustworthy Mach. Learn.*, 2023, pp. 225–253.
- [409] Y. Mo, D. Wu, Y. Wang, Y. Guo, and Y. Wang, "When adversarial training meets vision transformers: Recipes from training to architecture," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 18599–18611.
- [410] Y. Gao et al., "On the effectiveness of adversarial training against backdoor attacks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 10, pp. 14878–14888, Oct. 2024.
- [411] F. Croce et al., "RobustBench: A standardized adversarial robustness benchmark," *Adv. Neural Inf. Process. Syst.*, 2020.
- [412] A. E. Cinà et al., "AttackBench: Evaluating gradient-based attacks for adversarial examples," in *Proc. AAAI Conf. Artif. Intell.*, 2025, vol. 39, no. 3, pp. 2600–2608.
- [413] Y. Dong et al., "Benchmarking adversarial robustness on image classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 321–331.
- [414] A. Schwarzschild, M. Goldblum, A. Gupta, J. P. Dickerson, and T. Goldstein, "Just how toxic is data poisoning? A unified benchmark for backdoor and data poisoning attacks," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 9389–9398.
- [415] B. Wu et al., "Backdoorbench: A comprehensive benchmark of backdoor learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 10546–10559.
- [416] M. Zheng, X. Yan, Z. Zhu, H. Chen, and B. Wu, "BlackboxBench: A comprehensive benchmark of black-box adversarial attacks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 9, pp. 7867–7885, Sep. 2025.
- [417] C.-H. Weng, Y.-T. Lee, and S.-H. B. Wu, "On the trade-off between adversarial and backdoor robustness," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 11973–11983.
- [418] Z. Niu, Q. Miao, R. Jin, and G. Hua, "Towards unified robustness against both backdoor and adversarial attacks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 7589–7605, Dec. 2024.
- [419] J. Gilmer, N. Ford, N. Carlini, and E. Cubuk, "Adversarial examples are a natural consequence of test error in noise," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 2280–2289.
- [420] K. Kireev, M. Andriushchenko, and N. Flammarion, "On the effectiveness of adversarial training against common corruptions," in *Proc. Uncertainty Artif. Intell.*, 2022, pp. 1012–1021.
- [421] X. Xu, J. Zhang, F. Liu, M. Sugiyama, and M. S. Kankanhalli, "Enhancing adversarial contrastive learning via adversarial invariant regularization," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 16783–16803.